

TOPOS-THEORETIC BACKGROUND

In this chapter we review the classical topos-theoretic preliminaries necessary for understanding the contents of the book, for the benefit of readers who are not familiar with topos theory. We shall only assume an acquaintance with the basic notions of category theory.

Besides the original work [3], where the theory of toposes was first introduced, classical books on the subject, to which we shall systematically refer in the course of the chapter, are, in increasing order of technical sophistication, [46], [63] and [55]. We shall in particular follow section D1 of [55] in our presentation of the basic notions and results of first-order categorical logic, which are mainly due to [64], in sections 1.2, 1.3 and 1.4.

The structure of this introductory chapter is as follows.

In section 1.1 we introduce the notion of Grothendieck topos, review the basic properties of categories of sheaves and discuss the fundamental equivalence between geometric morphisms and flat functors.

In section 1.2 we introduce the concept of first-order theory and the various deductive systems for fragments of first-order logic that we shall consider in the course of the book.

In section 1.3 we review the way in which first-order theories can be soundly interpreted in categories having ‘enough’ structure; in particular, we explain the sense in which a Grothendieck topos (or, more generally, an elementary topos) can be regarded as a mathematical universe in which one can consider models of first-order theories.

In section 1.4 we present the fundamental concept of syntactic category of a first-order theory and illustrate the classification of models through structure-preserving functors defined on such categories. Lastly, we use syntactic categories for establishing the soundness and completeness of the categorical semantics defined in section 1.3.

1.1 Grothendieck toposes

Before introducing the notion of Grothendieck topos, we need to talk about Grothendieck topologies on categories. Indeed, Grothendieck toposes will be defined in section 1.1.2 as categories of sheaves on a small category with respect to a Grothendieck topology on it.

1.1.1 The notion of site

The notion of Grothendieck topology represents a categorical abstraction of the classical topological notion of covering of an open set of a topological space by a family of smaller open subsets.

Recall that a category is said to be *cartesian* if it has finite limits. A functor between two cartesian categories is said to be *cartesian* if it preserves finite limits.

Definition 1.1.1 (a) A *sieve* on an object c of a category $\mathcal{C}$ is a family of arrows in $\mathcal{C}$ with codomain c such that for any $f \in S$, $f \circ g \in S$ for any arrow g composable with f in $\mathcal{C}$.

(b) A sieve S is said to be *generated* by a family $\mathcal{F}$ of arrows contained in it if every arrow in S factors through an arrow in $\mathcal{F}$.

(c) A *Grothendieck topology* on a category $\mathcal{C}$ is an assignment J sending any object c of $\mathcal{C}$ to a collection $J(c)$ of sieves on $\mathcal{C}$ in such a way that the following properties are satisfied:

- (Maximality axiom) For any object c of $\mathcal{C}$, the maximal sieve $M_c := \{f \mid \text{cod}(f) = c\}$ on c belongs to $J(c)$.
- (Pullback stability) For any arrow $f : d \rightarrow c$ in $\mathcal{C}$ and any sieve $S \in J(c)$, the sieve $f^*(S) = \{g : e \rightarrow d \mid f \circ g \in S\}$ belongs to $J(d)$.
- (Transitivity) For any sieve S in $\mathcal{C}$ on an object c and any $T \in J(c)$, if $f^*(S) \in J(\text{dom}(f))$ for all $f \in T$ then $S \in J(c)$.

A sieve on an object c of $\mathcal{C}$ is said to be *J -covering* (or a *J -cover*), for a Grothendieck topology J on $\mathcal{C}$, if it belongs to $J(c)$.

(d) A *site* is a pair $(\mathcal{C}, J)$ consisting of a category $\mathcal{C}$ and a Grothendieck topology J on $\mathcal{C}$; it is said to be *small* (resp. *essentially small*) if the category $\mathcal{C}$ is small (resp. essentially small).

(e) Given two cartesian sites (i.e. sites whose underlying categories are cartesian) $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$, a *morphism of sites* $F : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ is a cartesian functor $F : \mathcal{C} \rightarrow \mathcal{D}$ which is cover-preserving (in the sense that the image under F of every J -covering sieve generates a K -covering sieve).

Remark 1.1.2 One can define morphisms of general, i.e. non-necessarily cartesian, sites (cf. section 1 of Exposé I.iii [3]); however, this general notion is more involved and we shall not use it in the book, so we do not discuss it here.

The following notions will be important for illuminating the relationship between topological spaces and the toposes of sheaves on them.

Recall that a *lattice* is a partially ordered set in which every pair of elements a and b has an infimum $a \wedge b$ and a supremum $a \vee b$.

Definition 1.1.3 (a) A Heyting algebra is a lattice H with bottom and top elements (denoted respectively by 0 and 1) in which for any two elements $a, b \in H$ there exists an element $a \Rightarrow b$ satisfying the universal property that for any $c \in H$, $c \leq a \Rightarrow b$ if and only if $c \wedge a \leq b$. The pseudocomplement $\neg a$ of an element a of H is defined to be $a \Rightarrow 0$. A Heyting algebra is said to be *complete* if it possesses arbitrary suprema (equivalently, arbitrary infima).

- (b) A *frame* F is a partially ordered set with arbitrary joins (and meets), in which the distributivity law of arbitrary joins with respect to finite meets holds:

$$\left(\bigvee_{i \in I} a_i \right) \wedge b = \bigvee_{i \in I} a_i \wedge b.$$

Since, by the special adjoint functor theorem, the functor $b \wedge - : F \rightarrow F$ has a right adjoint $b \Rightarrow - : F \rightarrow F$ if and only if it preserves arbitrary joins, a frame is the same thing as a complete Heyting algebra (cf. Remark C1.1.2 [55]).

The category **Frm** of frames has as objects frames and as arrows the frame homomorphisms between them (i.e. the maps which preserve finite meets and arbitrary joins).

- (c) A *locale* is a frame, regarded as an object of the opposite of the category **Frm**.

The category **Loc** of locales is the opposite **Frm**^{op} of the category of frames.

For any topological space X , the lattice $\mathcal{O}(X)$ of its open sets is a locale. Conversely, with any locale F one can associate a topological space X_F , whose points are the frame homomorphisms $F \rightarrow \{0, 1\}$ and whose open sets are the subsets of frame homomorphisms $F \rightarrow \{0, 1\}$ which send a given element $f \in F$ to 1. In fact, the assignments $X \rightarrow \mathcal{O}(X)$ and $F \rightarrow X_f$ lift to an adjunction between the category **Top** of topological spaces and continuous maps and the category **Loc** = **Frm**^{op} of locales (cf. section C1.2 of [55]).

The following are notable examples of Grothendieck topologies:

Examples 1.1.4

- (a) The *trivial topology* T on a category $\mathcal{C}$ is the Grothendieck topology on $\mathcal{C}$ whose only covering sieves are the maximal ones.
- (b) The *dense* or *double negation topology* on a category $\mathcal{C}$ is the Grothendieck topology D on $\mathcal{C}$ whose covering sieves are exactly the stably non-empty ones (recall that a sieve S in $\mathcal{C}$ on an object c is said to be *stably non-empty* if $f^*(S) \neq \emptyset$ for all arrows f in $\mathcal{C}$ with codomain c).
- (c) If $\mathcal{C}$ satisfies the *right Ore condition* (that is, the property that every pair of arrows with common codomain can be completed to a commutative square), the dense topology on $\mathcal{C}$ specializes to the *atomic topology* J_{at} on $\mathcal{C}$, whose covering sieves are exactly the non-empty ones.
- (d) Given a topological space X , there is a natural topology $J_{\mathcal{O}(X)}$ on the category $\mathcal{O}(X)$ of its open sets (whose objects are the open sets U of X and whose arrows are the inclusions $V \subseteq U$ between them), whose covering sieves are precisely the ones generated by (small) *covering families* (i.e. families of inclusions $\{U_i \subseteq U \mid i \in I\}$ such that $\bigcup_{i \in I} U_i = U$).
- (e) More generally, given a locale L , regarded as a preorder category, the *canonical topology* J_L on L has as covering sieves $S \in J_L(l)$ the sieves S generated by families $\{l_i \leq l \mid i \in I\}$ such that $\bigvee_{i \in I} l_i = l$.

1.1.2 Sheaves on a site

The introduction of the concept of Grothendieck topos stemmed from the observation that many important properties of topological spaces, such as compactness and connectedness, admit reformulations as categorically invariant properties of the associated categories of sheaves of sets; moreover, if a topological space X is sufficiently well-behaved (technically speaking, sober, cf. Remark 1.1.28), it can be recovered from the associated category $\mathbf{Sh}(X)$, as well as from the frame $\mathcal{O}(X)$, up to homeomorphism. Replacing topological spaces with the corresponding categories of sheaves on them presents the advantage that the latter are very rich in terms of categorical structure and can be studied by using a variety of different invariants naturally defined on them, notably including cohomology groups.

In studying the assignment $X \rightarrow \mathbf{Sh}(X)$ one immediately realizes that the definition of $\mathbf{Sh}(X)$ starting from X does not require the consideration of the points of X but only of the open sets of X and of the classical notion of covering for families of open subsets U_i of a given open set U (defined by saying that a given family $\{U_i \subseteq U \mid i \in I\}$ covers U if $\bigcup_{i \in I} U_i = U$). Now, the open sets of X can be organized in the preorder category $\mathcal{O}(X)$, while the above-mentioned notion of covering can be formulated as an additional *datum* on this category.

Abstracting these two fundamental ingredients in the construction of categories of sheaves led Grothendieck to introduce the notion of site recalled in section 1.1.1. Sheaves on a general site $(\mathcal{C}, J)$ are defined in a formally analogous way to sheaves of sets on a topological space, as follows.

Definition 1.1.5 Let $(\mathcal{C}, J)$ be a site.

- (a) A *presheaf* on a category $\mathcal{C}$ is a functor $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$.
- (b) A *sheaf* on $(\mathcal{C}, J)$ (or *J-sheaf* on $\mathcal{C}$) is a presheaf $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ on $\mathcal{C}$ such that for every J -covering sieve $S \in J(c)$ and every family $\{x_f \in P(\text{dom}(f)) \mid f \in S\}$ such that $P(g)(x_f) = x_{f \circ g}$ for any $f \in S$ and any arrow g in $\mathcal{C}$ composable with f , there exists a unique element $x \in P(c)$ such that $x_f = P(f)(x)$ for all $f \in S$.
- (c) The category $\mathbf{Sh}(\mathcal{C}, J)$ of sheaves on $(\mathcal{C}, J)$ has as objects the sheaves on $(\mathcal{C}, J)$ and as arrows the natural transformations between them, regarded as functors $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$.
- (d) A *Grothendieck topos* is a category which is equivalent to a category $\mathbf{Sh}(\mathcal{C}, J)$ of sheaves on a small site $(\mathcal{C}, J)$.
- (e) A *site of definition* for a Grothendieck topos $\mathcal{E}$ is a small site $(\mathcal{C}, J)$ such that $\mathcal{E} \simeq \mathbf{Sh}(\mathcal{C}, J)$.
- (f) A family of arrows $\mathcal{F}$ with common codomain e in a category $\mathcal{E}$ is said to be *epimorphic* if for any arrows $f, g : e \rightarrow a$ in $\mathcal{E}$, $f \circ h = g \circ h$ for all $h \in \mathcal{F}$ implies $f = g$.
- (g) A family $\mathcal{C}$ of objects of a category $\mathcal{E}$ is said to be *separating* for $\mathcal{E}$ if for any object e of $\mathcal{E}$, the arrows from objects of $\mathcal{C}$ to e form an epimorphic family.

- (h) A Grothendieck topology J on a locally small category $\mathcal{C}$ is said to be *subcanonical* if every representable functor $\mathrm{Hom}_{\mathcal{C}}(-, c) : \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ is a J -sheaf.

Notice that for any subcanonical topology J on $\mathcal{C}$, the Yoneda embedding yields a full embedding of $\mathcal{C}$ into $\mathbf{Sh}(\mathcal{C}, J)$.

- (i) A *subterminal object* of a Grothendieck topos $\mathcal{E}$ is an object e of $\mathcal{E}$ such that, denoting by 1 ‘the’ terminal object of $\mathcal{E}$, the unique arrow $e \rightarrow 1$ in $\mathcal{E}$ is a monomorphism.

Examples 1.1.6

- (a) The category **Set** of sets and functions between them is a Grothendieck topos. It can be represented as the category of sheaves on the one-point topological space.
- (b) More generally, given a small category $\mathcal{C}$ with the trivial topology T on it, the T -sheaves on $\mathcal{C}$ are clearly just the presheaves on $\mathcal{C}$; the category $\mathbf{Sh}(\mathcal{C}, T)$ thus coincides with the category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ of functors $\mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ and natural transformations between them.
- (c) For any topological space X , the topos $\mathbf{Sh}(\mathcal{O}_X, J_{\mathcal{O}(X)})$, where $J_{\mathcal{O}(X)}$ is the Grothendieck topology defined in Example 1.1.4(d), coincides with the usual category $\mathbf{Sh}(X)$ of sheaves of sets on the space X .

- (d) For any locale L , the topos $\mathbf{Sh}(L, J_L)$ is often simply denoted by $\mathbf{Sh}(L)$. The locale L can be recovered from it, up to isomorphism, as the frame $\mathrm{Sub}_{\mathbf{Sh}(L, J_L)}(1)$ of subterminal objects of $\mathbf{Sh}(L, J_L)$. A topos is said to be *localic* if it is (up to equivalence) of the form $\mathbf{Sh}(L)$ for a locale L .

More generally, for any *localic groupoid* $\mathcal{G}$ (i.e. groupoid internal to the category of locales, in the sense of section 1.3.3), there exists a Grothendieck topos $\mathbf{Sh}(\mathcal{G})$ of sheaves on it. By a well-known theorem of Joyal and Tierney (cf. [57]), every Grothendieck topos can be represented in this form.

- (e) Given a topological group G (with a prodiscrete topology), the category $\mathbf{Cont}(G)$ whose objects are the continuous (left) actions of G on discrete sets and whose arrows are the equivariant maps between them is a Grothendieck topos. In fact, $\mathbf{Cont}(G) \simeq \mathbf{Sh}(\mathcal{C}, J_{\mathrm{at}})$, where $\mathcal{C}$ is the full subcategory of $\mathbf{Cont}(G)$ on the non-empty transitive actions and J_{at} is the atomic topology on it.

More generally, for any *topological groupoid* $(d, c : G \rightarrow X)$ (i.e. groupoid internal to the category of topological spaces, in the sense of section 1.3.3), the category $\mathbf{Sh}_G(X)$ of G -equivariant sheaves on X is a Grothendieck topos. Butz and Moerdijk have shown in [11] that every Grothendieck topos with enough points can be represented in this form.

Definition 1.1.7 Let $(\mathcal{C}, J)$ be a site. A full subcategory $\mathcal{D}$ of $\mathcal{C}$ is said to be *J-dense* if for every object c of $\mathcal{C}$ the sieve generated by the family of arrows to c from objects in $\mathcal{D}$ is J -covering.

Given such a subcategory $\mathcal{D}$, the *Grothendieck topology* $J|_{\mathcal{D}}$ on $\mathcal{D}$ induced by J is defined as follows: for any sieve S in $\mathcal{D}$ on an object d , $S \in J|_{\mathcal{D}}(d)$ if and only if $\bar{S} \in J(d)$, where $\bar{S}$ is the sieve in $\mathcal{C}$ generated by the arrows in S .

A site of definition for a given topos can be seen as a sort of presentation of it by generators and relations (one can think of the objects of the category underlying the site as being the generators, and of the arrows and covering sieves as being the relations). This idea has for instance been exploited in section 8 of [16] for building various kinds of ordered structures presented by generators and relations.

As a group can have many different presentations, so a Grothendieck topos can have many different sites of definition, as for instance shown by the following theorem of [3] (cf. Theorem 4.1 in the Exposé I.iii):

Theorem 1.1.8 (Comparison Lemma). *Let $(\mathcal{C}, J)$ be a site such that $\mathcal{C}$ is locally small, and let $\mathcal{D}$ be a small full subcategory of $\mathcal{C}$ which is J -dense. Then the categories $\mathbf{Sh}(\mathcal{C}, J)$ and $\mathbf{Sh}(\mathcal{D}, J|_{\mathcal{D}})$ are equivalent.*

There are more refined versions of the Comparison Lemma in the literature; however, we shall not be concerned with them in this book. For a proof of the version reported above, we refer the reader to pp. 547-8 of [55].

Another important source of different sites of definition for a given topos is provided by its separating sets of objects.

Definition 1.1.9 The *canonical topology* $J_{\mathcal{E}}$ on a Grothendieck topos $\mathcal{E}$ is the Grothendieck topology on $\mathcal{E}$ whose covering sieves are exactly the sieves generated by small epimorphic families.

Every separating set of objects $\mathcal{C}$ of a Grothendieck topos $\mathcal{E}$, regarded as a full subcategory of $\mathcal{E}$, is $J_{\mathcal{E}}$ -dense and yields a representation $\mathcal{E} \simeq \mathbf{Sh}(\mathcal{C}, J_{\mathcal{E}}|_{\mathcal{C}})$ (cf. the proof of Theorem C2.2.8 [55]).

In spite of the existence of multiple representations for a given topos, there exists an intrinsic characterization, first obtained by Giraud, of the notion of Grothendieck topos: a locally small category is a Grothendieck topos if and only if it is a ∞ -pretopos with a separating set of objects (cf. Definition 1.3.17(b) below for the definition of an ∞ -pretopos). We shall not make use of this characterization in the book; rather, we shall systematically exploit the existence of different sites of definition for a given topos (cf. section 2.2).

1.1.3 Basic properties of categories of sheaves

Grothendieck toposes are very rich in terms of categorical structure: they possess all (small) limits and colimits, exponentials and a subobject classifier (as defined below in this section).

Recall that a *limit* (or *projective limit*, in the terminology of [3]) of a functor $D : \mathcal{I} \rightarrow \mathcal{E}$ defined on a small category $\mathcal{I}$ is a universal cone over D , i.e. it is an object e of $\mathcal{E}$ together with arrows $\lambda_i : e \rightarrow D(i)$ indexed by the objects of $\mathcal{I}$ such that, for any arrow $f : i \rightarrow j$ in $\mathcal{I}$, $D(f) \circ \lambda_i = \lambda_j$, which is universal with

this property. The notion of *colimit* (or *inductive limit*) is dual to that of limit; see, for instance, pp. 10-23 of [63] for more details. All the limits and colimits considered in this book will be *small*, i.e. indexed over small categories; so, we generically speak of (arbitrary) (co)limits, we actually mean *small (co)limits*.

An important property of functors which admit a right (resp. a left) adjoint, which will be exploited in the sequel, is the fact that they preserve all the colimits (resp. all the limits) which exist in the domain category.

For any small site $(\mathcal{C}, J)$ the inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ has a left adjoint $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$, called the *associated sheaf functor*, which preserves finite limits. Limits in $\mathbf{Sh}(\mathcal{C}, J)$ are computed pointwise as in $\mathbf{Set}$ (they are created by the inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$) while colimits in $\mathbf{Sh}(\mathcal{C}, J)$ can be calculated by taking the image under a_J of the colimit of the given diagram in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$.

To understand how exponentials are defined in general categories of sheaves, let us recall their construction in the base topos of sets.

For any two sets X and Y , we can always form the set Y^X of functions $X \rightarrow Y$. This set enjoys the following (universal) property in the category $\mathbf{Set}$ of sets: the familiar bijection

$$\text{Hom}_{\mathbf{Set}}(Z, Y^X) \cong \text{Hom}_{\mathbf{Set}}(Z \times X, Y)$$

is natural in both Y and Z and hence gives rise to an adjunction between the functor $- \times X : \mathbf{Set} \rightarrow \mathbf{Set}$ (left adjoint) and the functor $(-)^X : \mathbf{Set} \rightarrow \mathbf{Set}$ (right adjoint).

In a general category $\mathcal{C}$ with finite products, one defines the *exponential* A^B of two objects A and B by the universal property that for any object C of $\mathcal{C}$ the arrows $C \rightarrow A^B$ are in natural bijective correspondence with the arrows $C \times B \rightarrow A$. In a category $\mathbf{Sh}(\mathcal{C}, J)$ of sheaves on a small site $(\mathcal{C}, J)$ exponentials are computed as in $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$; specifically, an immediate application of the Yoneda lemma yields $P^Q(c) = \text{Nat}(yc \times Q, P)$ for all objects c of $\mathcal{C}$, where yc denotes the representable functor associated with the object c and Nat denotes the set of natural transformations.

Definition 1.1.10 A category $\mathcal{C}$ is said to be *cartesian closed* if it has finite products and exponentials for each pair of objects.

Note that a Heyting algebra can be described as a lattice with bottom and top elements which is cartesian closed when regarded as a preorder category with products.

Let us now recall the notion of *subobject classifier*. A *subobject* of an object c in a category $\mathcal{C}$ is an equivalence class of monomorphisms with codomain c modulo the equivalence relation which identifies two such monomorphisms when they factor one through the other. In the category $\mathbf{Set}$, subobjects of a set X correspond bijectively to subsets of X . The notion of subobject thus represents a natural categorical generalization of that of subset in set theory.

In the category **Set**, the subsets S of a given set X can be identified with their characteristic functions $\chi_S : X \rightarrow \{0, 1\}$; in fact, denoting by $\top : \{*\} = \mathbf{1}_{\mathbf{Set}} \rightarrow \{0, 1\}$ the function which sends $*$ to 1, we have a pullback square

$$\begin{array}{ccc} S & \xrightarrow{!} & \{*\} \\ \downarrow i & & \downarrow \top \\ X & \xrightarrow{\chi_S} & \{0, 1\}, \end{array}$$

where $i : S \rightarrow X$ is the inclusion and $! : S \rightarrow \{*\}$ is the unique arrow in **Set** from S to the terminal object $\mathbf{1}_{\mathbf{Set}} = \{*\}$.

This motivates the following

Definition 1.1.11 In a category $\mathcal{C}$ with finite limits a *subobject classifier* is a monomorphism $\top : \mathbf{1}_{\mathcal{C}} \rightarrow \Omega$, where $\mathbf{1}_{\mathcal{C}}$ is ‘the’ terminal object of $\mathcal{C}$, such that for every monomorphism $m : a' \rightarrow a$ there is a unique arrow $\chi_m : a \rightarrow \Omega$, called the *classifying arrow* of m , such that we have a pullback square

$$\begin{array}{ccc} a' & \xrightarrow{!} & \mathbf{1}_{\mathcal{C}} \\ \downarrow m & & \downarrow \top \\ a & \xrightarrow{\chi_m} & \Omega. \end{array}$$

Equivalently, if $\mathcal{C}$ is small, a subobject classifier is a representing object for the functor $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ sending an object a to the poset $\text{Sub}_{\mathcal{C}}(a)$ of subobjects of a and an arrow $f : a \rightarrow b$ to the pullback map $f^* : \text{Sub}_{\mathcal{C}}(b) \rightarrow \text{Sub}_{\mathcal{C}}(a)$.

Note that, for any object a of $\mathcal{C}$, we have an arrow $\in_a : a \times \Omega^a \rightarrow \Omega$ generalizing the belonging relation $\in$ of set theory, given by the transpose of the identity arrow on Ω^a .

Summarizing, we have the following result:

Theorem 1.1.12 *Let $(\mathcal{C}, J)$ be a small site.*

- (i) *The inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ has a left adjoint $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$, called the associated sheaf functor, which preserves finite limits.*
- (ii) *The category $\mathbf{Sh}(\mathcal{C}, J)$ has all (small) limits, which are preserved by the inclusion functor $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$; in particular, (small) limits are computed pointwise and the terminal object $\mathbf{1}_{\mathbf{Sh}(\mathcal{C}, J)}$ of $\mathbf{Sh}(\mathcal{C}, J)$ is the functor $T : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ sending each object $c \in \mathcal{C}$ to the singleton $\{*\}$.*
- (iii) *The associated sheaf functor $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ preserves (small) colimits; in particular, $\mathbf{Sh}(\mathcal{C}, J)$ has all (small) colimits.*
- (iv) *The category $\mathbf{Sh}(\mathcal{C}, J)$ has exponentials, which are constructed as in the topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$.*
- (v) *The category $\mathbf{Sh}(\mathcal{C}, J)$ has a subobject classifier.*

Subobject classifiers in Grothendieck toposes are constructed as follows:

- Given a site $(\mathcal{C}, J)$ and a sieve S in $\mathcal{C}$ on an object c , we say that S is *J-closed* if, for any arrow $f : d \rightarrow c$, $f^*(S) \in J(d)$ implies $f \in S$.
- For a small site $(\mathcal{C}, J)$, let us define $\Omega_J : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ as follows:

$$\Omega_J(c) = \{R \mid R \text{ is a } J\text{-closed sieve on } c\} \text{ (for an object } c \in \mathcal{C}\text{);}$$

$$\Omega_J(f) = f^*(-) \text{ (for an arrow } f \text{ in } \mathcal{C}\text{),}$$

where $f^*(-)$ denotes the operation of pullback of sieves in $\mathcal{C}$ along f .

Then the arrow $\top : 1_{\mathbf{Sh}(\mathcal{C}, J)} \rightarrow \Omega_J$ defined by

$$\top(*) (c) = M_c \text{ (the maximal sieve on } c\text{) for each } c \in \mathcal{C}$$

is a subobject classifier for $\mathbf{Sh}(\mathcal{C}, J)$.

- The classifying arrow $\chi_{A'} : A \rightarrow \Omega_J$ of a subobject $A' \subseteq A$ in $\mathbf{Sh}(\mathcal{C}, J)$ is given by

$$\chi_{A'}(c)(x) = \{f : d \rightarrow c \mid A(f)(x) \in A'(d)\}$$

for any $c \in \mathcal{C}$ and $x \in A(c)$.

For a detailed proof of this theorem the reader is referred to sections III.5-7 of [63].

The following result describes the main properties of subobject lattices in a Grothendieck topos:

Theorem 1.1.13

- (i) For any Grothendieck topos $\mathcal{E}$ and any object a of $\mathcal{E}$, the poset $\text{Sub}_{\mathcal{E}}(a)$ of all subobjects of a in $\mathcal{E}$ is a complete Heyting algebra.
- (ii) For any arrow $f : a \rightarrow b$ in a Grothendieck topos $\mathcal{E}$, the pullback functor $f^* : \text{Sub}_{\mathcal{E}}(b) \rightarrow \text{Sub}_{\mathcal{E}}(a)$ has both a left adjoint $\exists_f : \text{Sub}_{\mathcal{E}}(a) \rightarrow \text{Sub}_{\mathcal{E}}(b)$ and a right adjoint $\forall_f : \text{Sub}_{\mathcal{E}}(a) \rightarrow \text{Sub}_{\mathcal{E}}(b)$.

A detailed proof of this theorem can be found in section III.8 of [63].

1.1.4 Geometric morphisms

There are two natural types of morphisms to consider between toposes: *geometric morphisms* and *logical functors*. The former preserve the ‘geometric structure’ of toposes, the latter the ‘elementary logical’ one (cf. section 1.3.4). The latter are the natural class of morphisms to consider when one is interested in regarding an elementary topos as the syntactic category of a higher-order intuitionistic type theory (cf. section 1.3.4), while the former are the appropriate kind of morphisms to consider when toposes are regarded as generalized spaces or as classifying toposes of geometric theories (cf. section 2.1). Accordingly, we will focus in this book on the former class.

Definition 1.1.14 (a) A *geometric morphism* $f : \mathcal{F} \rightarrow \mathcal{E}$ of toposes is a pair of adjoint functors $f_* : \mathcal{F} \rightarrow \mathcal{E}$ and $f^* : \mathcal{E} \rightarrow \mathcal{F}$, respectively called the *direct*

image and the inverse image of f , such that the left adjoint f^* preserves finite limits (notice that f^* always preserves colimits as it has a right adjoint, while f_* always preserves limits).

- (b) A geometric transformation $\alpha : f \rightarrow g$ between two geometric morphisms $f, g : \mathcal{F} \rightarrow \mathcal{E}$ is a natural transformation $f^* \rightarrow g^*$ (equivalently, a natural transformation $g_* \rightarrow f_*$ – cf. the remarks following Definition A4.1.1 [55]).

We shall denote by $\mathfrak{B}\mathbf{top}$ the 2-category of Grothendieck toposes, geometric morphisms and geometric transformations between them. Given two Grothendieck toposes $\mathcal{E}$ and $\mathcal{F}$, we shall denote by $\mathbf{Geom}(\mathcal{E}, \mathcal{F})$ the category of geometric morphisms $\mathcal{E} \rightarrow \mathcal{F}$ and geometric transformations between them.

Recall that adjoints to a given functor, if they exist, are uniquely determined up to isomorphism. By the special adjoint functor theorem, giving a geometric morphism $\mathcal{F} \rightarrow \mathcal{E}$ is thus equivalent to giving a cocontinuous (i.e. colimit-preserving) cartesian functor $\mathcal{E} \rightarrow \mathcal{F}$.

Detailed proofs of the facts stated in the following list of examples can be found in sections VII.1-2 and VII.10 of [63].

Examples 1.1.15

- (a) Any continuous map $f : X \rightarrow Y$ of topological spaces induces a geometric morphism $\mathbf{Sh}(f) : \mathbf{Sh}(X) \rightarrow \mathbf{Sh}(Y)$ whose direct image $\mathbf{Sh}(f)_*$ is given by $\mathbf{Sh}(f)_*(P)(V) = P(f^{-1}(V))$ (for each $V \in \mathcal{O}(Y)$).
- (b) More generally, any arrow $f : L \rightarrow L'$ in $\mathbf{Loc}$ canonically induces a geometric morphism $\mathbf{Sh}(L) \rightarrow \mathbf{Sh}(L')$; conversely, every geometric morphism $\mathbf{Sh}(L) \rightarrow \mathbf{Sh}(L')$ is, up to isomorphism, of this form. This defines a full and faithful embedding of the (2-)category $\mathbf{Loc}$ into the (2-)category $\mathfrak{B}\mathbf{top}$.
- (c) Even more generally, any morphism of sites $f : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ induces a geometric morphism $\mathbf{Sh}(f) : \mathbf{Sh}(\mathcal{D}, K) \rightarrow \mathbf{Sh}(\mathcal{C}, J)$, whose direct image is the functor $- \circ f^{\mathrm{op}} : \mathbf{Sh}(\mathcal{D}, K) \rightarrow \mathbf{Sh}(\mathcal{C}, J)$.
- (d) Any functor $f : \mathcal{C} \rightarrow \mathcal{D}$ between small categories induces a geometric morphism $[f, \mathbf{Set}] : [\mathcal{C}, \mathbf{Set}] \rightarrow [\mathcal{D}, \mathbf{Set}]$, whose inverse image is given by $- \circ f : [\mathcal{D}, \mathbf{Set}] \rightarrow [\mathcal{C}, \mathbf{Set}]$. This functor has both a left and a right adjoint, respectively given by the left and right Kan extensions along the functor f . A geometric morphism whose inverse image has a left adjoint is said to be *essential*. If $\mathcal{D}$ is Cauchy-complete (i.e. all idempotents split in it) then every essential geometric morphism $[\mathcal{C}, \mathbf{Set}] \rightarrow [\mathcal{D}, \mathbf{Set}]$ is, up to isomorphism, of the form $[f, \mathbf{Set}]$ for a functor $f : \mathcal{C} \rightarrow \mathcal{D}$ (cf. Lemma A4.1.5 [55]).
- (e) For any small site $(\mathcal{C}, J)$, the canonical inclusion functor $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the direct image of a geometric morphism $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ whose inverse image $a_J : [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ is the associated sheaf functor.
- (f) For any (Grothendieck) topos $\mathcal{E}$ and any arrow $f : F \rightarrow E$ in $\mathcal{E}$, the pullback functor $f^* : \mathcal{E}/E \rightarrow \mathcal{E}/F$ between the slice toposes $\mathcal{E}/E$ and $\mathcal{E}/F$ is the inverse image of a geometric morphism $\mathcal{E}/F \rightarrow \mathcal{E}/E$.
- (g) Every Grothendieck topos $\mathcal{E}$ admits a unique (up to isomorphism) geometric morphism $\gamma_{\mathcal{E}} : \mathcal{E} \rightarrow \mathbf{Set}$. The direct image of $\gamma_{\mathcal{E}}$ is the *global sections functor*

$\mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, -) : \mathcal{E} \rightarrow \mathbf{Set}$, while the inverse image functor $\gamma_{\mathcal{E}}^*$ is given by $S \rightarrow \coprod_{s \in S} 1_{\mathcal{E}}$.

- Definition 1.1.16** (a) A geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$ is said to be a *surjection* if its inverse image $f^* : \mathcal{E} \rightarrow \mathcal{F}$ is faithful (equivalently, conservative).
 (b) A geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$ is said to be an *inclusion* if its direct image functor $f_* : \mathcal{F} \rightarrow \mathcal{E}$ is full and faithful.

Remark 1.1.17 As we shall see in section 1.3.4 (cf. Theorem 1.3.35), the *subtoposes* of $\mathcal{E}$, i.e. the (isomorphism classes of) geometric inclusions with codomain $\mathcal{E}$, correspond precisely to the local operators on $\mathcal{E}$.

Theorem 1.1.18 (cf. Theorems VII.4.6 and VII.4.8 [63]). *Every geometric morphism of (Grothendieck) toposes can be factored, in a unique way up to commuting isomorphisms, as a surjection followed by an inclusion.*

Definition 1.1.19 By a *point* of a topos $\mathcal{E}$, we mean a geometric morphism $\mathbf{Set} \rightarrow \mathcal{E}$.

Examples 1.1.20

- (a) For any essentially small site $(\mathcal{C}, J)$, the points of the topos $\mathbf{Sh}(\mathcal{C}, J)$ can be identified with the J -continuous flat functors $\mathcal{C} \rightarrow \mathbf{Set}$ (cf. Theorem 1.1.34).
 (b) For any locale L , the points of the topos $\mathbf{Sh}(L)$ correspond precisely to the frame homomorphisms $L \rightarrow \{0, 1\}$.
 (c) For any small category $\mathcal{C}$ and any object c of $\mathcal{C}$ we have a point $\mathbf{Set} \rightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ of the topos $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ whose inverse image is the evaluation functor at c .

Proposition 1.1.21 *Any collection of points P of a Grothendieck topos $\mathcal{E}$ indexed by a set X via a function $\xi : X \rightarrow P$ can be identified with a geometric morphism $\tilde{\xi} : [X, \mathbf{Set}] \rightarrow \mathcal{E}$ (where X is regarded as a discrete category).*

- Definition 1.1.22** (a) Let $\mathcal{E}$ be a topos and P a collection of points of $\mathcal{E}$. We say that P is *separating* for $\mathcal{E}$ if the points in P are jointly surjective, i.e. if the inverse image functors of the geometric morphisms in P jointly reflect isomorphisms (equivalently, if P is indexed by a set X via a function $\xi : X \rightarrow P$ as in Proposition 1.1.21, if the geometric morphism $\tilde{\xi} : [X, \mathbf{Set}] \rightarrow \mathcal{E}$ is surjective).
 (b) A topos is said to *have enough points* if the collection of all the points of $\mathcal{E}$ is separating for $\mathcal{E}$.

The following classical construction provides a way of endowing a given set of points of a topos with a natural topology:

Definition 1.1.23 Let $\xi : X \rightarrow P$ be an indexing of a set P of points of a Grothendieck topos $\mathcal{E}$ by a set X . The *subterminal topology* $\tau_\xi^\mathcal{E}$ on X is the image of the function $\phi_\mathcal{E} : \text{Sub}_\mathcal{E}(1) \rightarrow \mathcal{P}(X)$ given by

$$\phi_\mathcal{E}(u) = \{x \in X \mid \xi(x)^*(u) \cong 1_{\mathbf{Set}}\}.$$

We denote the space X endowed with the topology $\tau_\xi^\mathcal{E}$ by $X_{\tau_\xi^\mathcal{E}}$.

The following proposition follows at once from the definition of $\tau_\xi^\mathcal{E}$:

Proposition 1.1.24 *If P is a separating set of points for $\mathcal{E}$ then the frame $\mathcal{O}(X_{\tau_\xi^\mathcal{E}})$ of open sets of $X_{\tau_\xi^\mathcal{E}}$ is isomorphic to $\text{Sub}_\mathcal{E}(1)$ (via $\phi_\mathcal{E}$).*

This result was used in [16] and [26] to build spectra for various kinds of partially ordered structures and, combined with other results, Stone-type and Priestley-type dualities for them.

Remark 1.1.25 As observed in Remark C2.3.21 [55], for any small site $(\mathcal{C}, J)$, the subterminal objects of the topos $\mathbf{Sh}(\mathcal{C}, J)$ can be identified with the *J-ideals* on $\mathcal{C}$, i.e. with the collections I of objects of $\mathcal{C}$ with the property that for any arrow $f : c \rightarrow d$ in $\mathcal{C}$, $d \in I$ implies $c \in I$, and for any J -covering sieve S on an object c , if $\text{dom}(f) \in I$ for all $f \in S$ then $c \in I$.

Definition 1.1.26 A topological space is said to be *sober* if every irreducible closed subset is the closure of a unique point.

Example 1.1.27 Affine algebraic varieties (with the Zariski topology) are sober spaces.

Remark 1.1.28 The sober topological spaces are exactly the topological spaces X such that the canonical map $X \rightarrow X_{\tau_{\mathbf{Sh}(X)}^\mathcal{E}}$ is a homeomorphism; in other words, they are the spaces of points of localic toposes (up to homeomorphism).

1.1.5 Diaconescu's equivalence

The following theorem, already proved in [3], is of fundamental importance in topos theory and is commonly referred to in the literature as Diaconescu's equivalence in honour of R. Diaconescu who proved its relativization to an arbitrary base topos. Before stating it, we need to introduce the equally fundamental notion of flat functor.

Recall that the *category of elements* $\int P$ of a presheaf $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ has as objects the pairs (c, x) where c is an object of $\mathcal{C}$ and x is an element of $P(c)$ and as arrows $(c, x) \rightarrow (d, y)$ the arrows $f : c \rightarrow d$ in $\mathcal{C}$ such that $P(f)(y) = x$.

The following proposition exhibits a very general hom-tensor adjunction:

Proposition 1.1.29 *Let $\mathcal{C}$ be a small category and $\mathcal{E}$ a locally small cocomplete category. Then, for any functor $A : \mathcal{C} \rightarrow \mathcal{E}$, the functor $R_A : \mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ defined for each $e \in \mathcal{E}$ and $c \in \mathcal{C}$ by*

$$R_A(e)(c) = \text{Hom}_\mathcal{E}(A(c), e)$$

has a left adjoint $- \otimes_\mathcal{C} A : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathcal{E}$.

Sketch of proof The left adjoint $- \otimes_{\mathcal{C}} A$ sends a presheaf $P : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ to the ‘generalized tensor product’ $P \otimes_{\mathcal{C}} A = \text{colim}(A \circ \pi_P)$, where $\pi_P : \int P \rightarrow \mathcal{C}$ is the canonical projection to $\mathcal{C}$ from the category of elements $\int P$ of P . For more details, see the proof of Theorem I.5.2 [63]. $\square$

Definition 1.1.30 (a) A functor $A : \mathcal{C} \rightarrow \mathcal{E}$ from a small category $\mathcal{C}$ to a Grothendieck topos $\mathcal{E}$ is said to be *flat* if the functor $- \otimes_{\mathcal{C}} A : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathcal{E}$ preserves finite limits.
 (b) The full subcategory of $[\mathcal{C}, \mathcal{E}]$ on the flat functors will be denoted by $\mathbf{Flat}(\mathcal{C}, \mathcal{E})$.

Theorem 1.1.31 *Let $\mathcal{C}$ be a small category and $\mathcal{E}$ be a Grothendieck topos. Then we have an equivalence of categories*

$$\mathbf{Geom}(\mathcal{E}, [\mathcal{C}^{\text{op}}, \mathbf{Set}]) \simeq \mathbf{Flat}(\mathcal{C}, \mathcal{E})$$

natural in $\mathcal{E}$, which sends

- *a flat functor $A : \mathcal{C} \rightarrow \mathcal{E}$ to the geometric morphism $\mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ given by the functors R_A and $- \otimes_{\mathcal{C}} A$, and*
- *a geometric morphism $f : \mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ to the flat functor given by the composite $f^* \circ y$ of $f^* : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathcal{E}$ with the Yoneda embedding $y : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$.*

Sketch of proof By definition of a flat functor, the functor $- \otimes_{\mathcal{C}} A$ is the inverse image functor of a geometric morphism $\mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$. Conversely, given a geometric morphism $f : \mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$, the composite of f^* with the Yoneda embedding $y : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ yields a functor which is flat since $f^* \cong - \otimes_{\mathcal{C}} (f^* \circ y)$ (as the two functors take the same values on representables and both preserve colimits). The fact that $(- \otimes_{\mathcal{C}} A) \circ y \cong A$ follows immediately from the definition of $- \otimes_{\mathcal{C}} A$. For more details, see the proof of Corollary I.5.4 [63]. $\square$

Recall that a small category $\mathcal{C}$ is said to be *filtered* if it is non-empty, for any objects $c, d \in \mathcal{C}$ there exist an object $e \in \mathcal{C}$ and two arrows $f : c \rightarrow e$ and $g : d \rightarrow e$ and for any parallel arrows $f, g : a \rightarrow b$ in $\mathcal{C}$ there exists an arrow $h : b \rightarrow c$ in $\mathcal{C}$ such that $h \circ f = h \circ g$.

For a detailed proof of the facts stated in the following proposition we refer the reader to sections VII.7 and VII.9 of [63].

Proposition 1.1.32

- (i) *For any small category $\mathcal{C}$, a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is flat if and only if it is a filtered colimit of representables, equivalently if and only if its category of elements $\int F$ is filtered. More generally, a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{E}$ with values in a Grothendieck topos $\mathcal{E}$ is flat if and only if it is filtering i.e. satisfies the following conditions:*
 - (a) *For any object E of $\mathcal{E}$ there exist an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object b_i of $\mathcal{C}$ and a generalized element $E_i \rightarrow F(b_i)$ in $\mathcal{E}$.*

- (b) For any two objects c and d in $\mathcal{C}$ and any generalized element $\langle x, y \rangle : E \rightarrow F(c) \times F(d)$ in $\mathcal{E}$ there is an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object b_i of $\mathcal{C}$ with arrows $u_i : c \rightarrow b_i$ and $v_i : d \rightarrow b_i$ in $\mathcal{C}$ and a generalized element $z_i : E_i \rightarrow F(b_i)$ in $\mathcal{E}$ such that $\langle F(u_i), F(v_i) \rangle \circ z_i = \langle x, y \rangle \circ e_i$ for all $i \in I$.
- (c) For any two parallel arrows $u, v : d \rightarrow c$ in $\mathcal{C}$ and any generalized element $x : E \rightarrow F(c)$ in $\mathcal{E}$ for which $F(u) \circ x = F(v) \circ x$, there is an epimorphic family $\{e_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an arrow $w_i : c \rightarrow b_i$ and a generalized element $y_i : E_i \rightarrow F(b_i)$ such that $w_i \circ u = w_i \circ v$ and $F(w_i) \circ y_i = x \circ e_i$ for all $i \in I$.
- (ii) Let $\mathcal{C}$ be a category with finite limits and $\mathcal{E}$ a Grothendieck topos. Then a functor $\mathcal{C} \rightarrow \mathcal{E}$ is flat if and only if it preserves finite limits.

Definition 1.1.33 Let $\mathcal{E}$ be a Grothendieck topos and $(\mathcal{C}, J)$ a site. A functor $F : \mathcal{C} \rightarrow \mathcal{E}$ is said to be J -continuous if it sends J -covering sieves to epimorphic families.

The full subcategory of $\mathbf{Flat}(\mathcal{C}, \mathcal{E})$ on the J -continuous flat functors will be denoted by $\mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$.

Theorem 1.1.34 For any essentially small site $(\mathcal{C}, J)$ and Grothendieck topos $\mathcal{E}$, the above equivalence between geometric morphisms and flat functors restricts to an equivalence of categories

$$\mathbf{Geom}(\mathcal{E}, \mathbf{Sh}(\mathcal{C}, J)) \simeq \mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$$

natural in $\mathcal{E}$.

Sketch of proof Appeal to Theorem 1.1.31

- identifying the geometric morphisms $\mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ with the geometric morphisms $\mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ which factor through the canonical geometric inclusion $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$, and
- using the characterization of such morphisms as the geometric morphisms $f : \mathcal{E} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ such that the composite $f^* \circ y$ of the inverse image functor f^* of f with the Yoneda embedding $y : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ sends J -covering sieves to epimorphic families in $\mathcal{E}$.

For more details, see section VII.7 of [63]. □

1.2 First-order logic

In logic, *first-order languages* are a wide class of formal languages used for describing mathematical structures. The attribute ‘first-order’ means that all the universal and existential quantifications occurring in the axioms of a theory in such a language concern only individuals rather than collections of individuals (or other higher-order constructions on individuals). For instance, the property of a group to be abelian can be expressed in a first-order way in the language of

groups by the formula $(\forall x)(\forall y)(x + y = y + x)$, while the property of the ordered set $\mathbb{R}$ of real numbers that every bounded subset of it admits a supremum is not expressible in a first-order way (in the language of ordered sets) since it involves a quantification over subsets of the given structure rather than over elements of it (the formula defining it is a second-order one).

A first-order language contains *sorts*, which are meant to represent different *kinds* of individuals, *terms*, which denote individuals, and *formulae*, which make assertions about the individuals. Compound formulae are formed by using various logical operators, that is, either connectives (such as $\wedge$, $\vee$, $\Rightarrow$ etc.) or quantifiers ($\exists$ and $\forall$). For example, as we shall see in Example 1.2.7, it is natural to axiomatize the notion of (small) category by using a language with two sorts, one for the objects and one for the arrows.

It is well known, at least since the work of A. Tarski, that first-order languages can always be interpreted in the context of (a given model of) set theory (sorts are interpreted as sets, function symbols as functions and relation symbols as subsets). We will see in section 1.3 that one can meaningfully interpret them also in a general category, provided that the latter possesses enough categorical structure: sorts will be interpreted as *objects* of the given category, terms as *arrows* and formulae as *subobjects*, in a way which reflects their logical structure.

1.2.1 First-order theories

Definition 1.2.1 A first-order *signature* Σ consists of:

- (a) A set Σ -Sort of *sorts*.
- (b) A set Σ -Fun of *function symbols* f , each of which equipped with a finite list of source sorts and a single output sort: we write

$$f : A_1 \cdots A_n \rightarrow B$$

to indicate that f has source sorts $A_1, \dots, A_n$ and output sort B . If $n = 0$, f is called a *constant* of sort B . The *arity* of f is the number n of its source sorts.

- (c) A set Σ -Rel of *relation symbols* R , each of which equipped with a finite list of sorts: we write

$$R \mapsto A_1 \cdots A_n$$

to indicate that the list of sorts associated with R is $A_1, \dots, A_n$. The *arity* of R is the number n of sorts associated with it.

We shall say that a signature Σ is *finite* if it has only a finite number of sorts, function and relation symbols.

For each sort A of a signature Σ we assume to have at our disposal a stock of variables of sort A , used to denote individuals of type A .

A variable is said to be *free* in a given formula if it appears non-quantified in it, and *bound* otherwise. For example, in the formula $(\exists x)(x + y = 0)$ the variable y is free while the variable x is not. Note that a formula makes assertions

only about the non-quantified variables occurring in it. For this reason, formulae without free variables are called *sentences*, as their validity does not depend on the value of any variable.

Starting from variables, terms are built up by repeated ‘applications’ of function symbols to them, as follows:

Definition 1.2.2 Let Σ be a signature. The collection of *terms* over Σ is defined inductively, together with the specification for each term of its sort (we write $t : A$ or t^A to denote that t is a term of sort A), by means of the following clauses:

- (a) Any variable x^A of sort A is a term of sort A .
- (b) For any list of terms $t_1 : A_1, \dots, t_n : A_n$ and any function symbol $f : A_1 \cdots A_n \rightarrow B$, $f(t_1, \dots, t_n)$ is a term of sort B .

Starting from terms, one can build classes F of formulae inductively by means of the following formation rules, which simultaneously define the set $\text{FV}(\phi)$ of free variables of any formula ϕ in F :

- (i) *Relations*: $R(t_1, \dots, t_n)$ is in F for any terms $t_1 : A_1, \dots, t_n : A_n$ and any relation symbol $R \mapsto A_1 \cdots A_n$; $\text{FV}(R(t_1, \dots, t_n))$ is the set of variables occurring in some t_i (for $i = 1, \dots, n$).
- (ii) *Equality*: $(s = t)$ is in F for any terms s and t of the same sort; $\text{FV}(s = t)$ is the set of variables occurring in s or t (or both).
- (iii) *Truth*: $\top$ is in F ; $\text{FV}(\top) = \emptyset$.
- (iv) *Binary conjunction*: $(\phi \wedge \psi)$ is in F whenever ϕ and ψ are in F ; $\text{FV}(\phi \wedge \psi) = \text{FV}(\phi) \cup \text{FV}(\psi)$.
- (v) *Falsity*: $\perp$ is in F ; $\text{FV}(\perp) = \emptyset$.
- (vi) *Binary disjunction*: $(\phi \vee \psi)$ is in F whenever ϕ and ψ are in F ; $\text{FV}(\phi \vee \psi) = \text{FV}(\phi) \cup \text{FV}(\psi)$.
- (vii) *Implication*: $(\phi \Rightarrow \psi)$ is in F whenever ϕ and ψ are in F ; $\text{FV}(\phi \Rightarrow \psi) = \text{FV}(\phi) \cup \text{FV}(\psi)$.
- (viii) *Negation*: $\neg\phi$ is in F whenever ϕ is in F ; $\text{FV}(\neg\phi) = \text{FV}(\phi)$.
- (ix) *Existential quantification*: $(\exists x)\phi$ is in F whenever ϕ is in F and x is a free variable in ϕ ; $\text{FV}((\exists x)\phi) = \text{FV}(\phi) \setminus \{x\}$.
- (x) *Universal quantification*: $(\forall x)\phi$ is in F whenever ϕ is in F and x is a free variable in ϕ ; $\text{FV}((\forall x)\phi) = \text{FV}(\phi) \setminus \{x\}$.
- (xi) *Infinitary disjunction*: $\bigvee_{i \in I} \phi_i$ is in F whenever I is a set, ϕ_i is in F for each $i \in I$ and $\text{FV}(\bigvee_{i \in I} \phi_i) := \bigcup_{i \in I} \text{FV}(\phi_i)$ is finite.
- (xii) *Infinitary conjunction*: $\bigwedge_{i \in I} \phi_i$ is in F whenever I is a set, ϕ_i is in F for each $i \in I$ and $\text{FV}(\bigwedge_{i \in I} \phi_i) := \bigcup_{i \in I} \text{FV}(\phi_i)$ is finite.

We can think of $\top$ as an empty conjunction and of $\perp$ as an empty disjunction (cf. the deduction rules for conjunction and disjunction in section 1.2.2 below).

A *context* is a finite list $\vec{x} = (x_1, \dots, x_n)$ of distinct variables (the empty context, for $n = 0$, is allowed and indicated by []). We will often consider *formulae-in-context*, that is, formulae ϕ equipped with a context $\vec{x}$ such that all the free variables of ϕ occur in $\vec{x}$; we shall write either $\phi(\vec{x})$ or $\{\vec{x}. \phi\}$ for a formula ϕ in a context $\vec{x}$. The *canonical context* of a given formula is the context consisting of all the variables which appear freely in the formula.

Definition 1.2.3 In relation to the above-mentioned formation rules:

- (a) The set of *atomic formulae* over Σ is the smallest set closed under *Relations* and *Equality*.
- (b) The set of *Horn formulae* over Σ is the smallest set containing the class of atomic formulae and closed under *Truth* and *Binary conjunction*.
- (c) The set of *regular formulae* over Σ is the smallest set containing the class of atomic formulae and closed under *Truth*, *Binary conjunction* and *Existential quantification*.
- (d) The set of *coherent formulae* over Σ is the smallest set containing the class of atomic formulae and closed under *Truth*, *Binary conjunction*, *Existential quantification*, *False* and *Binary disjunction*.
- (e) The set of *first-order formulae* over Σ is the smallest set closed under all the above-mentioned formation rules except for the infinitary ones.
- (f) The *class of geometric formulae* over Σ is the smallest class containing the class of atomic formulae and closed under *Truth*, *Binary conjunction*, *Existential quantification*, *False* and *Infinitary disjunction*.
- (g) The *class of infinitary first-order formulae* over Σ is the smallest class closed under all the above-mentioned formation rules.

Definition 1.2.4 (a) By a *sequent* over a signature Σ we mean an expression of the form $(\phi \vdash_{\vec{x}} \psi)$, where ϕ and ψ are formulae over Σ in a context $\vec{x}$.
 (b) We say that a sequent $(\phi \vdash_{\vec{x}} \psi)$ is *Horn* (resp. *regular*, *coherent*, ...) if both ϕ and ψ are *Horn* (resp. *regular*, *coherent*, ...) formulae.

The intended meaning of a sequent $(\phi \vdash_{\vec{x}} \psi)$ is that ψ is a logical consequence of ϕ in the context $\vec{x}$, i.e. that for any ‘values’ taken by the variables in $\vec{x}$, if ϕ is satisfied by them then ψ is satisfied by them as well.

Notice that sequents are not necessary in the context of full first-order logic; indeed, any sequent $(\phi \vdash_{\vec{x}} \psi)$ has the same intended meaning as the formula $(\forall \vec{x})(\phi \Rightarrow \psi)$.

Definition 1.2.5 (a) By a (first-order) *theory* over a signature Σ , we mean a set $\mathbb{T}$ of sequents over Σ , whose elements are called the (non-logical) *axioms* of $\mathbb{T}$.

- (b) We say that $\mathbb{T}$ is an *algebraic theory* if its signature Σ has no relation symbols (apart from the equality relations on its sorts) and its axioms are all of the form $(\top \vdash_{\vec{x}} \phi)$ where ϕ is an atomic formula of the form $(s = t)$ and $\vec{x}$ its canonical context.

- (c) We say that $\mathbb{T}$ is a *Horn* (resp. *regular*, *coherent*, *geometric*) theory if all the sequents in $\mathbb{T}$ are Horn (resp. regular, coherent, geometric).
- (d) We say that $\mathbb{T}$ is a *universal Horn* theory if its axioms are all of the form $(\phi \vdash_{\vec{x}} \psi)$, where ϕ is a finite (possibly empty, i.e. equal to $\top$) conjunction of atomic formulae and ψ is an atomic formula or the formula $\perp$.
- (e) We say that a regular theory $\mathbb{T}$ is *cartesian* if its axioms can be well-ordered in such a way that each axiom is cartesian relative to the subtheory consisting of all the axioms preceding it in the ordering, in the sense that all the existential quantifications which appear in the axiom are provably unique relative to that subtheory (cf. section 1.2.2 for the notion of provability in fragments of first-order logic). A *cartesian formula* relative to a cartesian theory $\mathbb{T}$ over a signature Σ ($\mathbb{T}$ -*cartesian* formula, for short) is a formula built up from atomic formulae over Σ by only using binary conjunctions (possibly including the empty conjunction, i.e. $\top$) and $\mathbb{T}$ -provably unique existential quantifications. We say a sequent $(\phi \vdash_{\vec{x}} \psi)$ is cartesian relative to a cartesian theory $\mathbb{T}$ over a signature Σ ($\mathbb{T}$ -cartesian for short) if both ϕ and ψ are cartesian formulae over Σ (relative to $\mathbb{T}$).
- (f) We say that $\mathbb{T}$ is a *propositional theory* if its signature has no sorts, i.e. it only consists of 0-ary relation symbols.
- (g) We say that $\mathbb{T}$ is the *empty theory* over Σ if it does not contain any (non-logical) axioms.

Remark 1.2.6 The study of the model theory of algebraic theories has been initiated in the context of universal algebra and developed in a categorical setting since the pioneering work [59] by W. Lawvere introducing the functorial semantics of algebraic theories.

Example 1.2.7 An important example of a cartesian theory is given by the theory $\mathbb{C}$ of *small categories*. The language of $\mathbb{C}$ consists of two sorts O and A , respectively for objects and arrows, two function symbols $\text{dom}, \text{cod} : A \rightarrow O$ formalizing domain and codomain, a function symbol $1 : O \rightarrow A$ formalizing the map assigning to any object the identity arrow on it and a ternary predicate C formalizing the composition of arrows in the sense that $C(f, g, h)$ if and only if $h = f \circ g$ (notice that we have to use a relation symbol rather than a function symbol for formalizing composition since the latter is not everywhere defined). Over this signature, the axioms of $\mathbb{C}$ are the obvious ones. In particular, we have an axiom $(\text{dom}(f) = \text{cod}(g) \vdash_{f,g} (\exists h) C(f, g, h))$ expressing the existence of the composite of two arrows such that the codomain of the first coincides with the domain of the second, which is cartesian relative to the sequent $(C(f, g, h) \wedge C(f, g, h') \vdash_{f,g,h,h'} h = h')$ expressing the functionality of the predicate C .

Definition 1.2.8 Let $\mathbb{T}$ be a geometric theory over a signature Σ . The *cartesianization* (or *finite-limit part*) $\mathbb{T}_c$ of $\mathbb{T}$ is the cartesian theory over Σ consisting of all the $\mathbb{T}$ -cartesian sequents over Σ which are provable in $\mathbb{T}$.

1.2.2 Deduction systems for first-order logic

With each of the fragments of first-order logic introduced above, we can naturally associate a *deduction system*, in the same spirit as in classical first-order logic. Such systems will be formulated as *sequent calculi*, that is, they will consist of *inference rules* enabling one to derive a sequent from a collection of other sequents; we shall write

$$\frac{\Gamma}{\sigma}$$

to mean that the sequent σ can be inferred from a collection of sequents Γ . A double line between two sequents will mean that each of the sequents can be inferred from the other.

Given the axioms and inference rules below, the notion of *proof* (or *derivation*) is the usual one: a chain of inference rules whose premises are the axioms in the system and whose conclusion is the given sequent. Allowing the axioms of a theory $\mathbb{T}$ to be taken as premises yields the notion of *proof* (or *derivation*) *relative to a theory* $\mathbb{T}$.

Consider the following rules:

- The *rules for finite conjunction* are the axioms

$$(\phi \vdash_{\vec{x}} \top) \quad ((\phi \wedge \psi) \vdash_{\vec{x}} \phi) \quad ((\phi \wedge \psi) \vdash_{\vec{x}} \psi)$$

and the rule

$$\frac{(\phi \vdash_{\vec{x}} \psi)(\phi \vdash_{\vec{x}} \chi)}{(\phi \vdash_{\vec{x}} (\psi \wedge \chi))}.$$

- The *rules for finite disjunction* are the axioms

$$(\perp \vdash_{\vec{x}} \phi) \quad (\phi \vdash_{\vec{x}} (\phi \vee \psi)) \quad (\psi \vdash_{\vec{x}} \phi \vee \psi)$$

and the rule

$$\frac{(\phi \vdash_{\vec{x}} \chi)(\psi \vdash_{\vec{x}} \chi)}{((\phi \vee \psi) \vdash_{\vec{x}} \chi)}.$$

- The *rules for infinitary conjunction (resp. disjunction)* are the infinitary analogues of the rules for finite conjunction (resp. disjunction).
- The *rules for implication* consist of the double rule

$$\frac{(\phi \wedge \psi \vdash_{\vec{x}} \chi)}{(\psi \vdash_{\vec{x}} (\phi \Rightarrow \chi))}.$$

- The *rules for existential quantification* consist of the double rule

$$\frac{(\phi \vdash_{\vec{x}, y} \psi)}{((\exists y)\phi \vdash_{\vec{x}} \psi)},$$

where we assume that y is not free in ψ .

- The *rules for universal quantification* consist of the double rule

$$\frac{(\phi \vdash_{\vec{x}, y} \psi)}{(\phi \vdash_{\vec{x}} (\forall y)\psi)}.$$

- The *distributive axiom* is

$$(\phi \wedge (\psi \vee \chi)) \vdash_{\vec{x}} (\phi \wedge \psi) \vee (\phi \wedge \chi).$$

- The *Frobenius axiom* is

$$(\phi \wedge (\exists y)\psi) \vdash_{\vec{x}} (\exists y)(\phi \wedge \psi),$$

where y is a variable not in the context $\vec{x}$.

- The *Law of Excluded Middle* is

$$(\top \vdash_{\vec{x}} \phi \vee \neg\phi).$$

Remark 1.2.9 Note that these rules, although having been conceived with a particular semantics in mind (namely, the one that we shall describe in section 1.3), are completely formal: they do not have any meaning by themselves. Indeed, a fundamental principle of modern logic is that of strictly separating syntax and semantics, distinguishing in particular the notion of *provability* in a given formal system (intended as the possibility of deriving an assertion from a given set of premises by repeatedly applying certain specified ‘inference rules’) from that of *validity* (or satisfaction) in a given structure. Of central importance is therefore the investigation of the connections between syntax and semantics, possibly leading to soundness and completeness theorems relating the notion of provability in a deductive system with the notion of validity in an appropriately chosen class of structures. Results of this kind for the fragments of logic that we shall consider will be discussed in section 1.4.5.

1.2.3 Fragments of first-order logic

Definition 1.2.10 In addition to the usual structural rules of sequent calculi (*Identity axiom*, *Equality rules*, *Substitution rule* and *Cut rule*, cf. Definition D1.3.1 [55] for more details), our deduction systems consist of the following rules:

<i>Algebraic logic</i>	No additional rule
<i>Horn logic</i>	Finite conjunction
<i>Regular logic</i>	Finite conjunction, existential quantification and Frobenius axiom
<i>Coherent logic</i>	Finite conjunction, finite disjunction, existential quantification, distributive axiom and Frobenius axiom
<i>Geometric logic</i>	Finite conjunction, infinitary disjunction, existential quantification, ‘infinitary’ distributive axiom, Frobenius axiom
<i>Intuitionistic first-order logic</i>	All the finitary rules except for the law of excluded middle
<i>Classical first-order logic</i>	All the finitary rules

Definition 1.2.11 We say a sequent σ is *provable* (or *derivable*) in an algebraic (regular, coherent, ...) theory $\mathbb{T}$ ($\mathbb{T}$ -*provable*, for short) if there exists a derivation of σ relative to $\mathbb{T}$, in the appropriate fragment of first-order logic.

Remark 1.2.12 Whilst there is a *class* of geometric formulae over a given signature, there is only a *set* of provable-equivalence (with respect to geometric logic) classes of geometric formulae in a given context (cf. the proof of Lemma D1.4.10(iv) [55]). So one can define a geometric theory over a given signature Σ also by giving a class, rather than a set, of geometric sequents over Σ ; indeed, we are assured that there exists a set of geometric sequents that can act as representatives of the sequents in the given class in the sense that each of these sequents is provably equivalent in geometric logic to one of the sequents in the set. We shall specify geometric theories in this way at many points of the book.

In geometric logic, intuitionistic and classical provability of geometric sequents coincide, as shown by the following result:

Theorem 1.2.13 (cf. Proposition D3.1.16 [55]). *If a geometric sequent σ is derivable from the axioms of a geometric theory $\mathbb{T}$ using ‘classical geometric logic’ (i.e. the rules of geometric logic plus the Law of Excluded Middle), then σ is also derivable without using the Law of Excluded Middle.*

Remark 1.2.14 Since a given theory can in general be regarded as belonging to more than one of the fragments of geometric logic considered above, it is natural to wonder whether the notion of provability of sequents over the signature of the theory does not depend on the fragment. A positive answer to this question will be obtained in section 2.2 as an application of the ‘bridge’ technique.

1.3 Categorical semantics

Generalizing the classical Tarskian definition of satisfaction of first-order formulae in ordinary set-valued structures, one can naturally obtain, given a signature Σ , a notion of Σ -structure in a category with finite products, and define, according to the categorical structure present on the category, a notion of interpretation of an appropriate fragment of first-order logic into it.

Specifically, we shall introduce various classes of ‘logical’ categories, each of them providing a semantics for a corresponding fragment of first-order logic:

<i>Categories with finite products</i>	Algebraic logic
<i>Cartesian categories</i>	Cartesian logic
<i>Regular categories</i>	Regular logic
<i>Coherent categories</i>	Coherent logic
<i>Geometric categories</i>	Geometric logic
<i>Heyting categories</i>	First-order intuitionistic logic
<i>Boolean coherent categories</i>	First-order classical logic

The process by which these classes of categories are defined is quite canonical: first one looks at the set-theoretic structure needed to interpret the connectives and quantifiers occurring in the given fragment, then one proceeds to characterize it in categorical terms, and finally one uses the resulting categorical structure for interpreting the given fragment of logic. For example, one immediately realizes that in set theory the conjunction $\phi \wedge \psi$ of two formulae-in-context ϕ and ψ is interpreted as the intersection of the interpretations of ϕ and ψ . Now, intersections of subsets can be characterized in categorical terms as pullbacks of subobjects; this ensures that, in any category $\mathcal{C}$ with finite limits, it is possible to give a natural meaning to the conjunction of two formulae which are interpretable in $\mathcal{C}$.

Let us start defining the notion of Σ -structure in a category with finite products.

Definition 1.3.1 Let $\mathcal{C}$ be a category with finite products and Σ a signature. A Σ -structure M in $\mathcal{C}$ consists of:

- (a) An object MA of $\mathcal{C}$ for each sort A in Σ -Sort.
- (b) An arrow $Mf : MA_1 \times \cdots \times MA_n \rightarrow MB$ in $\mathcal{C}$ for each function symbol $f : A_1 \cdots A_n \rightarrow B$ in Σ -Fun.
- (c) A subobject $MR \rightrightarrows MA_1 \times \cdots \times MA_n$ in $\mathcal{C}$ for each relation symbol $R \rightrightarrows A_1 \cdots A_n$ in Σ -Rel.

Notice that if $n = 0$ then the cartesian product $MA_1 \times \cdots \times MA_n$ is equal to the terminal object $1_{\mathcal{C}}$ of $\mathcal{C}$.

Definition 1.3.2 A Σ -structure homomorphism $h : M \rightarrow N$ between two Σ -structures M and N in $\mathcal{C}$ is a collection of arrows $h_A : MA \rightarrow NA$ in $\mathcal{C}$ indexed by the sorts of Σ such that:

- (a) For each function symbol $f : A_1 \cdots A_n \rightarrow B$ in Σ -Fun, the diagram

$$\begin{array}{ccc}
 MA_1 \times \cdots \times MA_n & \xrightarrow{Mf} & MB \\
 \downarrow h_{A_1} \times \cdots \times h_{A_n} & & \downarrow h_B \\
 NA_1 \times \cdots \times NA_n & \xrightarrow{Nf} & NB
 \end{array}$$

commutes.

- (b) For each relation symbol $R \mapsto A_1 \cdots A_n$ in $\Sigma\text{-Rel}$, there is a commutative diagram in $\mathcal{C}$ of the form

$$\begin{array}{ccc} MR & \xrightarrow{\quad} & MA_1 \times \cdots \times MA_n \\ \downarrow & & \downarrow h_{A_1} \times \cdots \times h_{A_n} \\ NR & \xrightarrow{\quad} & NA_1 \times \cdots \times NA_n. \end{array}$$

Given a category $\mathcal{C}$ with finite products, Σ -structures in $\mathcal{C}$ and Σ -homomorphisms between them form a category, denoted by $\Sigma\text{-str}(\mathcal{C})$. Identities and composition in $\Sigma\text{-str}(\mathcal{C})$ are defined componentwise from those in $\mathcal{C}$.

Remark 1.3.3 If $\mathcal{C}$ and $\mathcal{D}$ are two categories with finite products then every functor $F : \mathcal{C} \rightarrow \mathcal{D}$ which preserves finite products and monomorphisms induces a functor $\Sigma\text{-str}(F) : \Sigma\text{-str}(\mathcal{C}) \rightarrow \Sigma\text{-str}(\mathcal{D})$ in the obvious way.

1.3.1 Classes of ‘logical’ categories

In this section we shall introduce classes of categories in which the fragments of first-order logic considered in section 1.2.3 can be naturally interpreted.

In any category $\mathcal{C}$ with pullbacks, pullbacks of monomorphisms are again monomorphisms; thus, for any arrow $f : a \rightarrow b$ in $\mathcal{C}$, we have a *pullback functor*

$$f^* : \text{Sub}_{\mathcal{C}}(b) \rightarrow \text{Sub}_{\mathcal{C}}(a).$$

Recall that by a *finite limit* in a category $\mathcal{C}$ we mean a limit of a functor $F : \mathcal{J} \rightarrow \mathcal{C}$ where $\mathcal{J}$ is a *finite category* (i.e. a category with only a finite number of objects and arrows).

Definition 1.3.4 A *cartesian* category is any category with finite limits.

As we shall see below, in cartesian categories we can interpret atomic formulae as well as finite conjunctions of them; in fact, conjunctions will be interpreted as pullbacks (i.e. intersections) of subobjects.

Definition 1.3.5 (a) Given two monomorphisms $m_1 : a_1 \rightarrow c$ and $m_2 : a_2 \rightarrow c$ of an object c in a category $\mathcal{C}$, we say that m_1 *factors through* m_2 if there is a (necessarily unique) arrow $r : a_1 \rightarrow a_2$ in $\mathcal{C}$ such that $m_2 \circ r = m_1$. Note that this defines a preorder relation $\leq$ on the collection $\text{Sub}_{\mathcal{C}}(c)$ of subobjects of a given object c .

(b) We say that a cartesian category $\mathcal{C}$ has *images* if for any morphism f of $\mathcal{C}$ there exists a subobject $\text{Im}(f)$ of its codomain, called the *image of f* , which is the least (with respect to the preorder $\leq$) subobject of $\text{cod}(f)$ through which f factors.

(c) A *regular category* is a cartesian category which has images that are stable under pullback.

Proposition 1.3.6 *Given an arrow $f : a \rightarrow b$ in a regular category $\mathcal{C}$, the pullback functor $f^* : \text{Sub}_{\mathcal{C}}(b) \rightarrow \text{Sub}_{\mathcal{C}}(a)$ has a left adjoint $\exists_f : \text{Sub}_{\mathcal{C}}(a) \rightarrow \text{Sub}_{\mathcal{C}}(b)$, which assigns to a subobject $m : c \rightarrow a$ the image of the composite arrow $f \circ m$.*

In a regular category, every arrow $f : a \rightarrow b$ factors uniquely through its image $\text{Im}(f) \rightarrow b$ as the composite $a \rightarrow \text{Im}(f) \rightarrow b$ of $\text{Im}(f) \rightarrow b$ with an arrow $c(f) : a \rightarrow \text{Im}(f)$; arrows of the form $c(f)$ for some f are called *covers*. In fact, every arrow in a regular category can be factored uniquely as a cover followed by a monomorphism, and covers are precisely the arrows g such that $\text{Im}(g) = 1_{\text{cod}(g)}$.

As we shall see in section 1.3.3, in regular categories we can interpret formulae built up from atomic formulae by using finite conjunctions and existential quantifications; in fact, the existential quantifications will be interpreted as images of certain arrows.

Definition 1.3.7 A *coherent category* is a regular category $\mathcal{C}$ in which every subobject lattice has finite unions which are stable under pullback.

As we shall see in section 1.3.3, in coherent categories we can interpret formulae built up from atomic formulae by using finite conjunctions, existential quantifications and finite disjunctions; in fact, finite disjunctions will be interpreted as finite unions of subobjects.

Note in passing that, if coproducts exist, a union of subobjects of an object c may be constructed as the image of the induced arrow from the coproduct of the domains of such subobjects to c .

Definition 1.3.8 (a) A (large) category $\mathcal{C}$ is said to be *well-powered* if each of the preorders $\text{Sub}_{\mathcal{C}}(a)$, $a \in \mathcal{C}$, is equivalent to a small category.

(b) A *geometric category* is a well-powered regular category whose subobject lattices have arbitrary unions which are stable under pullback.

As we shall see in section 1.3.3, in geometric categories we can interpret formulae built up from atomic formulae by using finite conjunctions, existential quantifications, and infinitary disjunctions; indeed, disjunctions will be interpreted as unions of subobjects.

To understand how to categorically interpret quantifiers, let us analyse their interpretations in the category **Set**.

Let X and Y be two sets. For any given subset $S \subseteq X \times Y$, we can consider the sets

$$\forall_p S := \{y \in Y \mid \text{for all } x \in X, (x, y) \in S\} \text{ and}$$

$$\exists_p S := \{y \in Y \mid \text{there exists } x \in X, (x, y) \in S\}.$$

The projection map $p : X \times Y \rightarrow Y$ induces a map at the level of powersets $p^* = p^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X \times Y)$. If we regard these powersets as poset categories (where the order is given by the inclusion relation) then this map becomes a

functor. Also, the assignments $S \rightarrow \forall_p S$ and $S \rightarrow \exists_p S$ yield functors $\forall_p, \exists_p : \mathcal{P}(X \times Y) \rightarrow \mathcal{P}(Y)$.

Proposition 1.3.9 *The functors $\exists_p$ and $\forall_p$ are respectively left and right adjoints to the functor $p^* : \mathcal{P}(Y) \rightarrow \mathcal{P}(X \times Y)$ which sends each subset $T \subseteq Y$ to its inverse image p^*T under p .*

Of course, the proposition generalizes to the case of an arbitrary function in place of the projection p . This motivates the following

Definition 1.3.10 A *Heyting category* is a coherent category $\mathcal{C}$ such that for any arrow $f : a \rightarrow b$ in $\mathcal{C}$, the pullback functor $f^* : \text{Sub}_{\mathcal{C}}(b) \rightarrow \text{Sub}_{\mathcal{C}}(a)$ has a right adjoint $\forall_f : \text{Sub}_{\mathcal{C}}(a) \rightarrow \text{Sub}_{\mathcal{C}}(b)$ (in addition to its left adjoint $\exists_f : \text{Sub}_{\mathcal{C}}(a) \rightarrow \text{Sub}_{\mathcal{C}}(b)$).

Proposition 1.3.11 (Lemma A1.4.13 [55]). *Let $a_1 \rhd a$ and $a_2 \rhd a$ be subobjects in a Heyting category. Then there exists a largest subobject $(a_1 \Rightarrow a_2) \rhd a$ such that $(a_1 \Rightarrow a_2) \cap a_1 \leq a_2$. Moreover, the binary operation on subobjects thus defined is stable under pullback.*

In particular, all the subobject lattices in a Heyting category are Heyting algebras.

The *Heyting negation* (also called *pseudocomplement*) $\neg a_1$ of a subobject $a_1 \rhd a$ is defined as $a_1 \Rightarrow 0_a$ (where 0_a is the zero subobject of a), and is characterized by the property of being the largest subobject of a disjoint from $a_1 \rhd a$. Notice that in general $a_1 \cup \neg a_1 \neq a$.

Proposition 1.3.11 ensures that in a Heyting category we may interpret full finitary first-order logic. In particular, the negation $\neg\phi(\vec{x})$ of a first-order formula-in-context $\phi(\vec{x})$ is interpreted as the Heyting pseudocomplement $\neg[[\vec{x}. \phi]]$. Note that, while it is always the case that $a \wedge \neg a = 0$ (in any Heyting algebra), $a \vee \neg a$ is in general different from 1, whence the Law of Excluded Middle is not sound with respect to general Heyting categories; indeed, the logic of Heyting categories is intuitionistic, not classical.

Definition 1.3.12 A coherent category $\mathcal{C}$ is said to be *Boolean* if every subobject $m : a \rhd c$ in $\mathcal{C}$ is complemented, in the sense that there exists a unique subobject n of c such that $m \cup n = 1_c$ and $m \cap n = 0_c$.

Since every Boolean algebra is a Heyting algebra, we have the following result:

Proposition 1.3.13 *Every Boolean coherent category is a Heyting category.*

Proposition 1.3.14 (cf. Lemma A1.4.18 [55]). *Every geometric category is a Heyting category.*

Proposition 1.3.15 *Every Grothendieck topos is a geometric category.*

Sketch of proof Well-poweredness immediately follows from the fact that by Giraud's theorem every Grothendieck topos has a separating set of objects, while the other properties easily follow from Theorems 1.1.12 and 1.1.13. $\square$

Thus every Grothendieck topos is a Heyting category. As we shall see in section 1.3.5, it is true more generally that every elementary topos is a Heyting category.

1.3.1.1 The internal language

Given a small category $\mathcal{C}$ with finite products one can define a first-order signature $\Sigma_{\mathcal{C}}$, called the *internal language* (or the *canonical signature*) of $\mathcal{C}$, for reasoning about $\mathcal{C}$ in a set-theoretic fashion.

Definition 1.3.16 The *internal language* $\Sigma_{\mathcal{C}}$ of $\mathcal{C}$ consists of

- one sort $\ulcorner A \urcorner$ for each object A of $\mathcal{C}$,
- one function symbol $\ulcorner f \urcorner : \ulcorner A_1 \urcorner \times \cdots \times \ulcorner A_n \urcorner \rightarrow \ulcorner B \urcorner$ for each arrow $f : A_1 \times \cdots \times A_n \rightarrow B$, and
- one relation symbol $\ulcorner R \urcorner \rightarrow \ulcorner A_1 \urcorner \cdots \ulcorner A_n \urcorner$ for each subobject $R \rightarrowtail A_1 \times \cdots \times A_n$ in $\mathcal{C}$.

We will sometimes write x^A instead of $x^{\ulcorner A \urcorner}$ to mean that x is a variable of sort $\ulcorner A \urcorner$.

In the sequel, we shall also informally consider the internal languages of non-small categories such as toposes, in situations where the size issues are not relevant.

There is a canonical $\Sigma_{\mathcal{C}}$ -structure $\mathcal{S}_{\mathcal{C}}$ in $\mathcal{C}$ called the *tautological $\Sigma_{\mathcal{C}}$ -structure*, which assigns A to $\ulcorner A \urcorner$, f to $\ulcorner f \urcorner$ and R to $\ulcorner R \urcorner$.

The usefulness of these notions lies in the fact that properties of $\mathcal{C}$ or constructions in it can often be formulated in terms of satisfaction of certain formulae over $\Sigma_{\mathcal{C}}$ in the tautological structure $\mathcal{S}_{\mathcal{C}}$. The internal language can thus be used for proving things about $\mathcal{C}$. Indeed, for any objects $A_1, \dots, A_n$ of $\mathcal{C}$ and any first-order formula $\phi(\vec{x})$ over $\Sigma_{\mathcal{C}}$, where $\vec{x} = (x_1^{\ulcorner A_1 \urcorner}, \dots, x_n^{\ulcorner A_n \urcorner})$, the set-theoretic expression $\{\vec{x} \in A_1 \times \cdots \times A_n \mid \phi(\vec{x})\}$ can be given a meaning, namely the interpretation of the formula $\phi(\vec{x})$ in the $\Sigma_{\mathcal{C}}$ -structure $\mathcal{S}_{\mathcal{C}}$.

We shall see basic examples of the use of the internal language of a topos in section 1.3.5.

It is also often convenient to use the reduced language $\Sigma_{\mathcal{C}}^f$ obtained from $\Sigma_{\mathcal{C}}$ by removing all the relation symbols.

1.3.2 Completions of ‘logical’ categories

It is important for many purposes to be able to complete the ‘logical’ categories that we have considered in section 1.3.1 with respect to certain kinds of colimits that they lack.

As far as it concerns regular categories, we can ‘complete’ them to regular categories in which quotients by equivalence relations always exist, by formally adding them. Such a construction is called *effectivization* and is characterized by the following universal property: for any regular category $\mathcal{C}$ there exists an effective regular category (i.e. a regular category in which every equivalence relation occurs as the kernel pair of some morphism) $\mathbf{Eff}(\mathcal{C})$ with a full and

faithful functor $i_{\mathbf{Eff}} : \mathcal{C} \rightarrow \mathbf{Eff}(\mathcal{C})$ such that for any effective regular category $\mathcal{D}$ the regular functors $\mathbf{Eff}(\mathcal{C}) \rightarrow \mathcal{D}$ correspond, naturally in $\mathcal{D}$, to the regular functors $\mathcal{C} \rightarrow \mathcal{D}$ (via composition with $i_{\mathbf{Eff}}$). For a detailed description of this construction the reader is referred to the proof of Corollary A3.3.10 [55].

Coherent (resp. geometric) categories do not possess in general finite (resp. arbitrary) coproducts, but it is possible to ‘complete’ them to coherent (resp. geometric) categories in which finite (resp. arbitrary) disjoint coproducts of objects exist. Such a construction is called *positivization* (resp. *infinitary positivization*) and is characterized by the following universal property:

- For any coherent category $\mathcal{C}$ there exists a positive coherent category (i.e. a coherent category having disjoint finite coproducts) $\mathbf{Pos}(\mathcal{C})$ with a full and faithful functor $i_{\mathbf{Pos}} : \mathcal{C} \rightarrow \mathbf{Pos}(\mathcal{C})$ such that for any positive category $\mathcal{D}$ the coherent functors $\mathbf{Pos}(\mathcal{C}) \rightarrow \mathcal{D}$ correspond, naturally in $\mathcal{D}$, to the coherent functors $\mathcal{C} \rightarrow \mathcal{D}$ (via composition with $i_{\mathbf{Pos}}$).
- For any geometric category $\mathcal{C}$ there exists an ∞ -positive coherent category (i.e. a geometric category which has disjoint arbitrary set-indexed coproducts) $\infty\text{-}\mathbf{Pos}(\mathcal{C})$ with a full and faithful functor $i_{\infty\text{-}\mathbf{Pos}} : \mathcal{C} \rightarrow \infty\text{-}\mathbf{Pos}(\mathcal{C})$ such that for any ∞ -positive category $\mathcal{D}$, the geometric functors $\infty\text{-}\mathbf{Pos}(\mathcal{C}) \rightarrow \mathcal{D}$ correspond, naturally in $\mathcal{D}$, to the geometric functors $\mathcal{C} \rightarrow \mathcal{D}$ (by composition with $i_{\infty\text{-}\mathbf{Pos}}$).

More details about these constructions can be found at pp. 34-35 of [55].

Definition 1.3.17 (a) A *pretopos* is a positive and effective coherent category.
 (b) An ∞ -*pretopos* is an ∞ -positive and effective geometric category.

Remarks 1.3.18 (a) For any coherent category $\mathcal{C}$, the category $\mathbf{Eff}(\mathbf{Pos}(\mathcal{C}))$, with the canonical embedding $i_{\mathbf{Pr}} : \mathcal{C} \hookrightarrow \mathbf{Eff}(\mathbf{Pos}(\mathcal{C}))$, satisfies the universal property of the *pretopos completion* $\mathcal{P}_{\mathcal{C}}$ of $\mathcal{C}$, i.e. $\mathbf{Eff}(\mathbf{Pos}(\mathcal{C}))$ is a pretopos such that for any pretopos $\mathcal{D}$ the coherent functors $\mathcal{P}_{\mathcal{C}} \rightarrow \mathcal{D}$ correspond precisely to the coherent functors $\mathcal{C} \rightarrow \mathcal{D}$ (via composition with $i_{\mathbf{Pr}}$).

(b) For any geometric category $\mathcal{C}$, the category $\mathbf{Eff}(\infty\text{-}\mathbf{Pos}(\mathcal{C}))$, with the canonical embedding $i_{\infty\text{-}\mathbf{Pr}} : \mathcal{C} \hookrightarrow \mathbf{Eff}(\infty\text{-}\mathbf{Pos}(\mathcal{C}))$, satisfies the universal property of the ∞ -*pretopos completion* $\mathcal{P}_{\mathcal{C}}^{\infty}$ of $\mathcal{C}$, i.e. $\mathbf{Eff}(\infty\text{-}\mathbf{Pos}(\mathcal{C}))$ is a ∞ -pretopos such that for any ∞ -pretopos $\mathcal{D}$ the geometric functors $\mathcal{P}_{\mathcal{C}}^{\infty} \rightarrow \mathcal{D}$ correspond precisely to the geometric functors $\mathcal{C} \rightarrow \mathcal{D}$ (via composition with $i_{\infty\text{-}\mathbf{Pr}}$).

1.3.3 Models of first-order theories in categories

In this section we shall define the notion of model of a theory belonging to a fragment of first-order logic in a category in which it is interpretable.

First-order terms over a given signature can be interpreted in any category with finite products.

Definition 1.3.19 Let M be a Σ -structure in a category $\mathcal{C}$ with finite products. Given a term-in-context $\{\vec{x}.t\}$ over Σ (where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$ and $t : B$, say), we inductively define an arrow

$$[[\vec{x}.t]]_M : MA_1 \times \cdots \times MA_n \rightarrow MB$$

in $\mathcal{C}$ as follows:

- If t is a variable then it is necessarily x_i for a unique $i \in \{1, \dots, n\}$, and we set $[[\vec{x}.t]]_M$ equal to the i th product projection

$$\pi_i : MA_1 \times \cdots \times MA_n \rightarrow MA_i.$$

- If t is $f(t_1, \dots, t_m)$ (where $t_i : C_i$, say), then we set $[[\vec{x}.t]]_M$ equal to the composite

$$MA_1 \times \cdots \times MA_n \xrightarrow{<[[\vec{x}.t_1]]_M, \dots, [[\vec{x}.t_m]]_M>} MC_1 \times \cdots \times MC_m \xrightarrow{Mf} MB$$

In order to interpret first-order formulae in a given category $\mathcal{C}$, we need to have a certain amount of categorical structure present on $\mathcal{C}$ so to give a meaning to the logical connectives and quantifiers which appear in the formulae. For example, to interpret finitary conjunctions, one needs to be able to form pullbacks, while to interpret disjunctions one needs to be able to take unions of subobjects, etc. In fact, the larger the fragment of logic is, the larger the amount of categorical structure is required to interpret it.

Let M be a Σ -structure in a category $\mathcal{C}$ with finite limits. A formula-in-context $\{\vec{x}.\phi\}$ over Σ (where $\vec{x} = x_1, \dots, x_n$ and $x_i : A_i$, say) will be interpreted as a subobject $[[\vec{x}.\phi]]_M \rightarrowtail MA_1 \times \cdots \times MA_n$ according to the following inductive rules:

- If $\phi(\vec{x})$ is $R(t_1, \dots, t_m)$ where R is a relation symbol (of type $B_1, \dots, B_m$, say), then $[[\vec{x}.\phi]]_M$ is the pullback

$$\begin{array}{ccc} [[\vec{x}.\phi]]_M & \xrightarrow{\quad} & MR \\ \downarrow & & \downarrow \\ MA_1 \times \cdots \times MA_n & \xrightarrow{<[[\vec{x}.t_1]]_M, \dots, [[\vec{x}.t_m]]_M>} & MB_1 \times \cdots \times MB_m. \end{array}$$

- If $\phi(\vec{x})$ is $(s = t)$, where s and t are terms of sort B , then $[[\vec{x}.\phi]]_M$ is the equalizer of $[[\vec{x}.s]]_M, [[\vec{x}.t]]_M : MA_1 \times \cdots \times MA_n \rightarrow MB$.
- If $\phi(\vec{x})$ is $\top$ then $[[\vec{x}.\phi]]_M$ is the top element of $\text{Sub}_{\mathcal{C}}(MA_1 \times \cdots \times MA_n)$.
- If ϕ is $\psi \wedge \chi$ then $[[\vec{x}.\phi]]_M$ is the intersection (i.e. pullback)

$$\begin{array}{ccc} [[\vec{x}.\phi]]_M & \xrightarrow{\quad} & [[\vec{x}.\chi]]_M \\ \downarrow & & \downarrow \\ [[\vec{x}.\psi]]_M & \xrightarrow{\quad} & MA_1 \times \cdots \times MA_n. \end{array}$$

- If $\phi(\vec{x})$ is $\perp$ and $\mathcal{C}$ is a coherent category then $[[\vec{x}.\phi]]_M$ is the bottom element of $\text{Sub}_{\mathcal{C}}(MA_1 \times \cdots \times MA_n)$.

- If ϕ is $\psi \vee \chi$ and $\mathcal{C}$ is a coherent category then $[[\vec{x}. \phi]]_M$ is the union of the subobjects $[[\vec{x}. \psi]]_M$ and $[[\vec{x}. \chi]]_M$.
- If ϕ is $\psi \Rightarrow \chi$ and $\mathcal{C}$ is a Heyting category then $[[\vec{x}. \phi]]_M$ is the implication $[[\vec{x}. \psi]]_M \Rightarrow [[\vec{x}. \chi]]_M$ in the Heyting algebra $\text{Sub}_{\mathcal{C}}(MA_1 \times \cdots \times MA_n)$ (similarly, the negation $\neg\psi$ is interpreted as the pseudocomplement of $[[\vec{x}. \psi]]_M$).
- If ϕ is $(\exists y)\psi$ where y is of sort B and $\mathcal{C}$ is a regular category then $[[\vec{x}. \phi]]_M$ is the image of the composite

$$[[\vec{x}, y. \psi]]_M \longrightarrow MA_1 \times \cdots \times MA_n \times MB \xrightarrow{\pi} MA_1 \times \cdots \times MA_n,$$

where π is the product projection on the first n factors.

- If ϕ is $(\forall y)\psi$ where y is of sort B and $\mathcal{C}$ is a Heyting category then $[[\vec{x}. \phi]]_M$ is $\forall_{\pi}([[\vec{x}, y. \psi]]_M)$, where π is the same projection as above.
- If ϕ is $\bigvee_{i \in I} \phi_i$ and $\mathcal{C}$ is a geometric category then $[[\vec{x}. \phi]]_M$ is the union of the subobjects $[[\vec{x}. \phi_i]]_M$.
- If ϕ is $\bigwedge_{i \in I} \phi_i$ and $\mathcal{C}$ has arbitrary intersections of subobjects then $[[\vec{x}. \phi]]_M$ is the intersection of the subobjects $[[\vec{x}. \phi_i]]_M$.

Definition 1.3.20 Let M be a Σ -structure in a category $\mathcal{C}$ with finite products.

- A sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ which is interpretable in $\mathcal{C}$ is said to be *satisfied in M* (and one writes $M \models \sigma$) if $[[\vec{x}. \phi]]_M \leq [[\vec{x}. \psi]]_M$ in $\text{Sub}_{\mathcal{C}}(MA_1 \times \cdots \times MA_n)$.
- If $\mathbb{T}$ is a theory over Σ whose axioms are interpretable in $\mathcal{C}$, M is said to be a *model* of $\mathbb{T}$ (or a $\mathbb{T}$ -*model*) if all the axioms of $\mathbb{T}$ are satisfied in M .
- M is said to be a *conservative* model of a theory $\mathbb{T}$ over Σ (within a given fragment of first-order logic) if any sequent over Σ which is satisfied in M is provable in $\mathbb{T}$ and conversely.

The full subcategory of $\Sigma\text{-str}(\mathcal{C})$ whose objects are the models of $\mathbb{T}$ will be denoted by $\mathbb{T}\text{-mod}(\mathcal{C})$.

Remark 1.3.21 One can easily prove, by induction on the structure of ϕ , that for any geometric formula $\phi(\vec{x}) = \phi(x_1^{A_1}, \dots, x_n^{A_n})$ over a signature Σ and any Σ -structure homomorphism $h : M \rightarrow N$, there is a commutative diagram in $\mathcal{C}$ of the form

$$\begin{array}{ccc} [[\vec{x}. \phi]]_M & \longrightarrow & MA_1 \times \cdots \times MA_n \\ \downarrow & & \downarrow h_{A_1} \times \cdots \times h_{A_n} \\ [[\vec{x}. \phi]]_N & \longrightarrow & NA_1 \times \cdots \times NA_n. \end{array}$$

This is not in general true if ϕ is a first-order non-geometric formula. This motivates the following

Definition 1.3.22 A homomorphism of models of a first-order theory $\mathbb{T}$ in a category $\mathcal{C}$ is said to be an *elementary morphism* if there is a commutative diagram as in Remark 1.3.21 for all the first-order formulae $\phi(\vec{x})$ over the signature of $\mathbb{T}$. The category of $\mathbb{T}$ -models in $\mathcal{C}$ and elementary embeddings between them will be denoted by $\mathbb{T}\text{-mod}_e(\mathcal{C})$.

If $\mathcal{C}$ and $\mathcal{D}$ are categories with finite products then any functor $F : \mathcal{C} \rightarrow \mathcal{D}$ which preserves finite products and monomorphisms canonically induces a functor $\Sigma\text{-str}(F) : \Sigma\text{-str}(\mathcal{C}) \rightarrow \Sigma\text{-str}(\mathcal{D})$ in the obvious way.

We say that a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between two cartesian (resp. regular, coherent, geometric, Heyting) categories is *cartesian* (resp. *regular*, *coherent*, *geometric*, *Heyting*) if it preserves finite limits (resp. finite limits and images, finite limits and images and finite unions of subobjects, finite limits and images and arbitrary unions of subobjects, finite limits and images and Heyting implications between subobjects).

Recall that a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be *conservative* if it reflects isomorphisms (i.e. for any arrow $f : c \rightarrow c'$ in $\mathcal{C}$, if $F(f)$ is an isomorphism in $\mathcal{D}$ then f is an isomorphism in $\mathcal{C}$).

Lemma 1.3.23 (Lemma D1.2.13 [55]). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a cartesian (resp. regular, coherent, geometric, Heyting) functor between categories of the appropriate kind, M a Σ -structure in $\mathcal{C}$ and σ a sequent over Σ interpretable in $\mathcal{C}$. If $M \models \sigma$ in $\mathcal{C}$ then $F(M) \models \sigma$ in $\mathcal{D}$. The converse implication holds if F is conservative.*

Proof An easy induction shows that F preserves the interpretations of all formulae-in-context interpretable in the appropriate class of categories; from this the first assertion immediately follows. To prove the second, it suffices to note that a sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ is satisfied in M if and only if the canonical subobject $[[\vec{x}. \phi \wedge \psi]]_M \rightarrowtail [[\vec{x}. \phi]]_M$ is an isomorphism. $\square$

Lemma 1.3.23 implies the following result:

Theorem 1.3.24 *If $\mathbb{T}$ is a regular (resp. coherent, ...) theory over Σ , then for any regular (resp. coherent, ...) functor $F : \mathcal{C} \rightarrow \mathcal{D}$ the functor $\Sigma\text{-str}(F) : \Sigma\text{-str}(\mathcal{C}) \rightarrow \Sigma\text{-str}(\mathcal{D})$ defined above restricts to a functor $\mathbb{T}\text{-mod}(F) : \mathbb{T}\text{-mod}(\mathcal{C}) \rightarrow \mathbb{T}\text{-mod}(\mathcal{D})$.*

Remark 1.3.25 We shall see in section 1.4.5, by using the concept of syntactic category, that the categorical semantics defined above is *sound* and *complete*, in the sense that the provability of a sequent in a given theory belonging to a fragment of first-order logic is equivalent to its validity in all the models of the theory in categories in which the sequent is interpretable.

The following list of examples shows that the notion of model of a first-order theory in a category captures several important notions of genuine mathematical interest:

Examples 1.3.26

- (a) A *topological group* can be seen as a model of the algebraic theory of groups in the category of topological spaces.
More generally, a *topological groupoid* can be seen as a model of the theory $\mathbb{G}$ of groupoids (obtained from the theory $\mathbb{C}$ of small categories considered in Example 1.2.7 by adding a unary function symbol Inv of sort A for formalizing the operation of inverse of an arrow and the obvious axioms for it) in the category of topological spaces.
- (b) Similarly, an *algebraic* (resp. *Lie*) *group* is a model of the algebraic theory of groups in the category of algebraic varieties (resp. in the category of smooth manifolds).
- (c) A *sheaf of rings* (more generally, a sheaf whose sections are models of a Horn theory $\mathbb{T}$) on a topological space X can be seen as a model of the theory of rings (resp. of the theory $\mathbb{T}$) in the topos $\mathbf{Sh}(X)$ of sheaves on X .
- (d) A *sheaf of local rings* on a topological space X (i.e. a sheaf of rings on X whose stalks are all local rings) is precisely a model of the theory of local rings in the topos $\mathbf{Sh}(X)$.
- (e) A *family of set-based models* of a geometric theory $\mathbb{T}$ indexed by a set I can be seen as a model of $\mathbb{T}$ in the functor category $[I, \mathbf{Set}]$.

Remark 1.3.27 An important aspect of categorical semantics is that a given piece of syntax can be soundly interpreted in a variety of independent contexts. The concrete results obtained in this way cannot in general be deduced from one another, but arise as different instances of a unique general result lying at the syntactic level. This can be useful in practice to avoid reproving the same results in different contexts, when it is possible to lift them to the syntactic level. For example, the fact that the category of algebraic groups and that of Lie groups have finite products are both immediate consequences of the syntactic property of the theory of groups to be algebraic.

1.3.4 Elementary toposes

The notion of elementary topos was introduced by W. Lawvere and M. Tierney in the late sixties as a generalization of the concept of Grothendieck topos in which it would still be possible to do some sort of abstract sheaf theory and consider models of arbitrary finitary first-order theories.

- Definition 1.3.28** (a) An *elementary topos* is a cartesian closed category (i.e. a category with finite products in which exponentials exist for any pair of objects) with finite limits and a subobject classifier.
- (b) A *logical functor* between elementary toposes is a functor preserving finite limits, exponentials and the subobject classifier.

As the name suggests, the notion of elementary topos can be formalized elementarily in the first-order language of categories (cf. Example 1.2.7).

By Theorem 1.1.12, every Grothendieck topos is an elementary topos. There are examples of elementary toposes that are not Grothendieck toposes, and which

are useful in connection with the logical study of higher-order intuitionistic type theories and realizability. In fact, elementary toposes are exactly the syntactic categories, with respect to logical functors, of higher-order intuitionistic type theories (cf. section D4.3 of [55] for more details).

As shown by Giraud's theorem, an essential feature which distinguishes Grothendieck toposes among general elementary toposes is the fact that they admit separating sets of objects (equivalently, (small) sites of definition). The presence of sites allows one to effectively use geometric intuition when dealing with toposes, and to study first-order mathematical theories of a general specified form (technically speaking, geometric theories, cf. section 2.1.1) through their classifying toposes. On the other hand, when studying aspects of toposes arising from their elementary categorical structure, it is often possible, and even natural, to establish the relevant results in the setting elementary toposes. We shall see various instances of this phenomenon in the book.

The name 'topos' for this kind of categories is justified by the fact that it is possible to lift many natural notions and constructions which apply to Grothendieck toposes to this level of generality. For instance, as we shall see, the notion of Grothendieck topology J on a category $\mathcal{C}$ corresponds to an invariant notion defined at the level of the presheaf topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$, namely to a *local operator* (also called *Lawvere-Tierney topology*) on it.

By an internal Heyting algebra in a topos $\mathcal{E}$ we mean a model of the algebraic theory of Heyting algebras in $\mathcal{E}$, in the sense of section 1.3.3, that is an object L of $\mathcal{E}$ with arrows $\wedge, \vee, \Rightarrow : L \times L \rightarrow L$ and $0, 1 : 1_{\mathcal{E}} \rightarrow L$ which make the diagrams expressing the identities used in the equational definition of a Heyting algebra commutative.

Theorem 1.3.29 *Any elementary topos $\mathcal{E}$ is a Heyting category and the subobject classifier Ω of $\mathcal{E}$ has the structure of an internal Heyting algebra, inducing by the Yoneda lemma a natural structure of Heyting algebra on each subobject lattice in $\mathcal{E}$.*

Sketch of proof Let $\mathcal{E}$ be an elementary topos. One can prove that $\mathcal{E}$ has all finite colimits (cf. section IV.5 of [63]). Using this, one can construct cover-mono factorizations for arrows which are stable under pullback, thereby showing that $\mathcal{E}$ is a regular category. The pullback functors between subobject lattices in $\mathcal{E}$ thus admit left adjoints. The fact that they also admit right adjoints follows from the existence of the cartesian closed structure (cf. the proof of Theorem I.9.4 [63]). The existence of these right adjoints ensures that finite unions of subobjects in $\mathcal{E}$, which exist in $\mathcal{E}$ since $\mathcal{E}$ has finite colimits as well as images, are stable under pullback. Given this, it is not hard to prove that Ω has the structure of an internal Heyting algebra. Specifically, one defines $0 : 1_{\mathcal{E}} \rightarrow \Omega$ to be the classifying arrow of the zero subobject $0 \rightarrow 1_{\mathcal{E}}$, $1 : 1_{\mathcal{E}} \rightarrow \Omega$ to be the arrow $\top, \wedge : \Omega \times \Omega \rightarrow \Omega$ to be the classifying arrow of the monomorphism $< \top, \top > : 1_{\mathcal{E}} \rightarrow \Omega \times \Omega$, $\Rightarrow : \Omega \times \Omega \rightarrow \Omega$ to be the classifying arrow of the equalizer of the arrows $\wedge$ and π_1 and $\vee : \Omega \times \Omega \rightarrow \Omega$ to be the classifying arrow

of the union of the two subobjects $\pi_1^*(\top)$ and $\pi_2^*(\top)$, where π_1 and π_2 are the two canonical projections $\Omega \times \Omega \rightarrow \Omega$ (for more details, see the proof of Lemma A1.6.3 [55]).

It is also immediate to see that the Heyting algebra structure on the subobject lattices in $\mathcal{E}$ is induced by this internal structure via the Yoneda lemma. $\square$

Definition 1.3.30 Let $\mathcal{E}$ be an elementary topos with subobject classifier $\top : 1 \rightarrow \Omega$. A *local operator* (or *Lawvere-Tierney topology*) on $\mathcal{E}$ is an arrow $j : \Omega \rightarrow \Omega$ in $\mathcal{E}$ such that the three diagrams

$$\begin{array}{ccc} 1 & & \Omega \\ \downarrow \top & \searrow \top & \downarrow j \\ \Omega & \xrightarrow{j} & \Omega \end{array} \quad \begin{array}{ccc} \Omega & & \Omega \\ \downarrow j & \searrow j & \downarrow j \\ \Omega & \xrightarrow{j} & \Omega \end{array} \quad \begin{array}{ccc} \Omega \times \Omega & \xrightarrow{\wedge} & \Omega \\ \downarrow j \times j & & \downarrow j \\ \Omega \times \Omega & \xrightarrow{\wedge} & \Omega \end{array}$$

commute (where $\wedge : \Omega \times \Omega \rightarrow \Omega$ is the meet operation of the internal Heyting algebra Ω).

Interestingly, the notion of local operator admits several equivalent characterizations.

- Definition 1.3.31** (a) A *closure operation* on subobjects in an elementary topos $\mathcal{E}$ is an operation c sending any subobject m in $\text{Sub}_{\mathcal{E}}(A)$ to a subobject $c(m)$ in $\text{Sub}_{\mathcal{E}}(A)$ in such a way that $m \leq c(m)$ and $c(c(m)) = c(m)$ for all m , where $\leq$ is the natural ordering between subobjects.
- (b) A *universal closure operation* on an elementary topos $\mathcal{E}$ is a closure operation c on subobjects which commutes with pullback (in particular, intersection) of subobjects.

An important example of a universal closure operation on a topos $\mathcal{E}$ is the *double negation* one, which assigns to any subobject m in $\mathcal{E}$ the subobject $\neg\neg m$ (where the Heyting pseudocomplement $\neg$ is taken in the relevant subobject lattice).

Theorem 1.3.32 For any elementary topos $\mathcal{E}$, there is a bijection between universal closure operations on $\mathcal{E}$ and local operators on $\mathcal{E}$.

Sketch of proof The bijection of the theorem sends a universal closure operation c on $\mathcal{E}$ to the local operator $j_c : \Omega \rightarrow \Omega$ given by the classifying map of the subobject $c(1 \xrightarrow{\top} \Omega)$, and a local operator j to the closure operation c_j induced by composing classifying arrows with j . For more details, see the proof of Lemma A4.4.2 [55]. $\square$

One can define an abstract notion of c -sheaf for a universal closure operation c on an elementary topos $\mathcal{E}$, as follows.

Definition 1.3.33 Let $\mathcal{E}$ be an elementary topos and c a universal closure operation on $\mathcal{E}$. An object A of $\mathcal{E}$ is said to be a *c -sheaf* if for any monomorphism

$m : B' \rightarrowtail B$ in $\mathcal{E}$ which is c -dense (i.e. such that $c(m) = 1_B$) and any arrow $f' : B' \rightarrow A$ in $\mathcal{E}$, there exists exactly one arrow $f : B \rightarrow A$ such that $f \circ m = f'$.

The full subcategory of $\mathcal{E}$ on the c -sheaves will be denoted by $\mathbf{sh}_c(\mathcal{E})$.

Definition 1.3.34 A *reflector* L on a topos $\mathcal{E}$ is the left adjoint functor of a reflection (i.e. full embedding admitting a left adjoint) of a full subcategory of $\mathcal{E}$ into $\mathcal{E}$. A reflector is said to be *cartesian* if it preserves finite limits.

For a proof of the following result the reader is referred to sections A4.3 and A4.4 of [55].

Theorem 1.3.35 *For any local operator j on an elementary (resp. Grothendieck) topos $\mathcal{E}$, $\mathbf{sh}_{c_j}(\mathcal{E})$ is an elementary (resp. Grothendieck) topos, and the inclusion $\mathbf{sh}_{c_j}(\mathcal{E}) \hookrightarrow \mathcal{E}$ has a left adjoint $a_j : \mathcal{E} \rightarrow \mathbf{sh}_{c_j}(\mathcal{E})$ which preserves finite limits. In fact, local operators on $\mathcal{E}$ correspond bijectively to (isomorphism classes of) geometric inclusions to $\mathcal{E}$, and to (isomorphism classes of) cartesian reflectors on $\mathcal{E}$.*

Sketch of proof A closure operation c_j can be recovered from the associated cartesian reflector L_j by means of the pullbacks

$$\begin{array}{ccc} c_j(A') & \longrightarrow & L_j A' \\ \downarrow c_j(m) & & \downarrow L_j m \\ A & \xrightarrow{\eta_A} & L_j A, \end{array}$$

where $m : A' \rightarrowtail A$ is an arbitrary monomorphism in $\mathcal{E}$ and η is the unit of the reflection induced by L_j . $\square$

Theorem 1.3.36 *Let $\mathcal{C}$ be a small category. Then the Grothendieck topologies J on $\mathcal{C}$ correspond exactly to the local operators on the presheaf topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$. In fact, if J is the Grothendieck topology corresponding to a local operator j , an object of $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is a J -sheaf (in the sense of Definition 1.1.5(b)) if and only if it is a c_j -sheaf (in the sense of Definition 1.3.33).*

Sketch of proof The correspondence of the theorem sends a local operator $j : \Omega \rightarrow \Omega$ to the subobject $J \rightarrowtail \Omega$ which it classifies, that is to the Grothendieck topology J on $\mathcal{C}$ defined by

$$S \in J(c) \text{ if and only if } j(c)(S) = M_c.$$

In the other direction, it sends a Grothendieck topology J , regarded as a subobject $J \rightarrowtail \Omega$, to the arrow $j : \Omega \rightarrow \Omega$ that classifies it.

For more details, see section V.4 of [63]. $\square$

1.3.5 Toposes as mathematical universes

We have seen in section 1.3.1.1 that, given a category $\mathcal{C}$ with finite products, there is a first-order signature $\Sigma_{\mathcal{C}}$, called the internal language of $\mathcal{C}$, which can be profitably used for reasoning about $\mathcal{C}$ in a set-theoretic fashion, that is by using ‘elements’.

If $\mathcal{C}$ is an elementary topos, we can extend the internal language by allowing the formation of formulae of the kind $\tau \in \Gamma$, where τ is a term of sort $\ulcorner A \urcorner$ and Γ is a term of sort $\ulcorner \Omega^A \urcorner$. Indeed, we may interpret such a formula as the subobject whose classifying arrow is the composite

$$W \xrightarrow{\langle \tau, \Gamma \rangle} A \times \Omega^A \xrightarrow{\in_A} \Omega,$$

where W is the product of (the objects representing the) sorts of the variables occurring either in τ or in Γ (considered without repetitions), $\langle \tau, \Gamma \rangle$ is the induced map to the product and $\in_A$ is the transpose of the identity arrow on Ω^A along the adjunction between $(-)^A$ and $A \times -$. Note that an object A of $\mathcal{C}$ gives rise to a constant term of type $\ulcorner \Omega^A \urcorner$. In fact, Ω^A behaves like the power set of A in $\mathcal{C}$.

We may also allow the formation of terms of type $t(w)$, where t is a term of type $\ulcorner B^A \urcorner$ and w is a term of type $\ulcorner A \urcorner$; indeed, we may interpret $t(w)$ as the composite $\text{ev}_{A,B} \circ \langle t, w \rangle$, where $\text{ev}_{A,B}$ is the evaluation arrow $B^A \times A \rightarrow B$.

Thus in a topos we can also interpret all the common formulae that we use in set theory.

Note that, the logic of a topos being at least intuitionistic (cf. Theorems 1.3.29 and 1.4.15), for any topos $\mathcal{E}$, any formal proof involving first-order sequents over the signature $\Sigma_{\mathcal{E}}$ will be sound when interpreted in the tautological structure $\mathcal{S}_{\mathcal{E}}$ provided that the law of excluded middle or any other non-constructive principles are not employed in it. In other words, the existence of the internal language of a topos and the corresponding tautological structure justifies the use of the standard set-theoretic intuition when working ‘inside’ a topos provided that one avoids invoking non-constructive principles.

An example of a reformulation of basic properties of sets in the internal language of a topos is provided by the following proposition.

Proposition 1.3.37 (cf. Lemma D1.3.11 [55]). *Let $\mathcal{E}$ be a topos. Then*

- (i) *An arrow $f : A \rightarrow A$ of $\mathcal{E}$ is the identity on A if and only if $(\top \vdash_x \ulcorner f \urcorner(x) = x)$ holds in $\mathcal{S}_{\mathcal{E}}$.*
- (ii) *An arrow $f : A \rightarrow C$ of $\mathcal{E}$ is the composite of two arrows $g : A \rightarrow B$ and $h : B \rightarrow C$ if and only if $(\top \vdash_x \ulcorner f \urcorner(x) = \ulcorner h \urcorner(\ulcorner g \urcorner(x)))$ holds in $\mathcal{S}_{\mathcal{E}}$.*
- (iii) *An arrow $f : A \rightarrow B$ of $\mathcal{E}$ is a monomorphism if and only if $(\ulcorner f \urcorner(x) = \ulcorner f \urcorner(x') \vdash_{x,x'} x = x')$ holds in $\mathcal{S}_{\mathcal{E}}$.*
- (iv) *An arrow $f : A \rightarrow B$ of $\mathcal{E}$ is an epimorphism if and only if $(\top \vdash_y (\exists x)(\ulcorner f \urcorner(x) = y))$ holds in $\mathcal{S}_{\mathcal{E}}$.*

- (v) An object A of $\mathcal{E}$ is terminal if and only if $(\top \vdash (\exists x)\top)$ and $(\top \vdash_{x,x'} (x = x'))$ hold in $\mathcal{S}_{\mathcal{E}}$ (where x and x' are variables of sort $\ulcorner A \urcorner$).

1.3.5.1 Validity in an elementary topos

Definition 1.3.38 We say that a first-order formula $\phi(\vec{x})$ over a signature Σ is *valid* in an elementary topos $\mathcal{E}$ if for every Σ -structure M in $\mathcal{E}$ the sequent $(\top \vdash_{\vec{x}} \phi)$ is satisfied in M .

Theorem 1.3.39 Let Σ be a signature and $\phi(\vec{x})$ a first-order formula over Σ . Then $\phi(\vec{x})$ is provable in intuitionistic (finitary) first-order logic if and only if it is valid in every elementary topos.

Sketch of proof The soundness part follows from Theorems 1.3.29 and 1.4.15. The completeness part follows from the existence of canonical Kripke models and the fact that, given a poset P and a Kripke model $\mathcal{U}$ on P , there is a model $\mathcal{U}^*$ in the topos $[P, \mathbf{Set}]$ such that the first-order sequents valid in $\mathcal{U}$ are exactly those which are valid in $\mathcal{U}^*$. For more details, see section 8.4 of [46]. $\square$

An elementary topos can thus be considered as a ‘mathematical universe’ in which one can do mathematics similarly to how one does it in the classical context of sets, with the only proviso being that one must argue constructively. This constructivity requirement should not be merely seen as a limitation, but rather as an opportunity for a more refined development of mathematics allowing one to discriminate between different notions that ‘collapse’ to equivalent ones in classical logic.

1.3.5.2 Kripke-Joyal semantics

Kripke-Joyal semantics represents the analogue for toposes of the usual Tarskian notion of satisfaction of a first-order formula by a tuple of elements of a set-theoretic structure in which the given formula is interpretable. It clearly makes no sense to speak of elements of a structure in a topos, but we can replace the classical notion of element of a set with that of *generalized element* of an object. A generalized element of an object B of a topos $\mathcal{E}$ is simply an arrow $\alpha : E \rightarrow B$ with codomain B . Note that, by the Yoneda lemma, an object is determined, up to isomorphism, by the collection of all its generalized elements (represented by the functor $\text{Hom}_{\mathcal{E}}(-, B)$).

Definition 1.3.40 Let $\mathcal{E}$ be an elementary topos and $\mathcal{S}_{\mathcal{E}}$ the tautological $\Sigma_{\mathcal{E}}$ -structure in $\mathcal{E}$. Given a first-order formula $\phi(x)$ in a variable x of sort $\ulcorner A \urcorner$ and a generalized element $\alpha : E \rightarrow A$ of A , we stipulate that

$$E \models \phi(\alpha) \text{ if and only if } \alpha \text{ factors through } [[x \cdot \phi]]_{\mathcal{S}_{\mathcal{E}}} \rightarrowtail A.$$

Theorem 1.3.41 Under the hypotheses of Definition 1.3.40, let $\alpha : E \rightarrow A$ be a generalized element of A and $\phi(x), \psi(x)$ formulae with a free variable x of sort $\ulcorner A \urcorner$. Then

- $E \models (\phi \wedge \psi)(\alpha)$ if and only if $E \models \phi(\alpha)$ and $E \models \psi(\alpha)$.
- $E \models (\phi \vee \psi)(\alpha)$ if and only if there are jointly epimorphic arrows $p : E' \rightarrow E$ and $q : E'' \rightarrow E$ in $\mathcal{E}$ such that both $E' \models \phi(\alpha \circ p)$ and $E'' \models \psi(\alpha \circ q)$.
- $E \models (\phi \Rightarrow \psi)(\alpha)$ if and only if, for any arrow $p : E' \rightarrow E$ of $\mathcal{E}$ such that $E' \models \phi(\alpha \circ p)$, $E' \models \psi(\alpha \circ p)$.
- $E \models (\neg\phi)(\alpha)$ if and only if, whenever $p : E' \rightarrow E$ is an arrow of $\mathcal{E}$ such that $E' \models \phi(\alpha \circ p)$, $E' \cong 0_{\mathcal{E}}$.

If $\phi(x, y)$ has an additional free variable y of sort $\ulcorner B \urcorner$ then

- $E \models (\exists y)\phi(\alpha, y)$ if and only if there exist an epimorphism $p : E' \rightarrow E$ and a generalized element $\beta : E' \rightarrow B$ such that $E' \models \phi(\alpha \circ p, \beta)$.
- $E \models (\forall y)\phi(\alpha, y)$ if and only if, for every object E' of $\mathcal{E}$, every arrow $p : E' \rightarrow E$ of $\mathcal{E}$ and every generalized element $\beta : E' \rightarrow B$, $E' \models \phi(\alpha \circ p, \beta)$.

Of course, both Definition 1.3.40 and Theorem 1.3.41 can be generalized to the case of formulae with an arbitrary (finite) number of free variables. For a proof of Theorem 1.3.41 we refer the reader to section 6.6 of [9].

Remark 1.3.42 If $\mathcal{E}$ is a Grothendieck topos then infinitary formulae can also be interpreted in it. The infinitary version of the second point of the theorem reads as follows: $E \models (\bigvee_{i \in I} \phi_i)(\alpha)$ if and only if there is an epimorphic family of arrows $\{p_i : E_i \rightarrow E \mid i \in I\}$ in $\mathcal{E}$ such that $E_i \models \phi(\alpha \circ p_i)$ for all $i \in I$.

1.4 Syntactic categories

In this section we shall introduce the fundamental notion of *syntactic category* of a theory within a given fragment of first-order logic.

1.4.1 Definition

Let $\phi(\vec{x})$ and $\psi(\vec{y})$ be two formulae-in-context over a first-order signature, where $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$ are contexts of the same type and length. We say that $\phi(\vec{x})$ and $\psi(\vec{y})$ are α -equivalent if $\psi(\vec{y})$ is obtained from $\phi(\vec{x})$ by (possibly renaming bound variables in ϕ and) replacing every free occurrence of x_i in ϕ by y_i . We write $\{\vec{x}. \phi\}$ for the α -equivalence class of the formula-in-context $\phi(\vec{x})$.

Definition 1.4.1 (a) Let $\mathbb{T}$ be a geometric theory over a signature Σ . The *syntactic category* $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$ has as objects the α -equivalence classes of geometric formulae-in-context $\{\vec{x}. \phi\}$ over Σ and as arrows $\{\vec{x}. \phi\} \rightarrow \{\vec{y}. \psi\}$ (where the contexts $\vec{x}$ and $\vec{y}$ are supposed to be disjoint without loss of generality) the $\mathbb{T}$ -provable-equivalence classes $[\theta]$ of geometric formulae $\theta(\vec{x}, \vec{y})$ which are $\mathbb{T}$ -provably functional, i.e. such that the sequents

$$\begin{aligned} &(\phi \vdash_{\vec{x}} (\exists \vec{y}) \theta), \\ &(\theta \vdash_{\vec{x}, \vec{y}} \phi \wedge \psi), \text{ and} \\ &((\theta \wedge \theta[\vec{z}/\vec{y}]) \vdash_{\vec{x}, \vec{y}, \vec{z}} (\vec{y} = \vec{z})) \end{aligned}$$

are provable in $\mathbb{T}$.

The composite of two arrows

$$\{\vec{x}.\phi\} \xrightarrow{[\theta]} \{\vec{y}.\psi\} \xrightarrow{[\gamma]} \{\vec{z}.\chi\}$$

is defined as the $\mathbb{T}$ -provable-equivalence class of the formula $(\exists \vec{y})(\theta \wedge \gamma)$. The identity arrow on an object $\{\vec{x}.\phi\}$ is the arrow

$$\{\vec{x}.\phi\} \xrightarrow{[\phi \wedge \vec{x}' = \vec{x}]} \{\vec{x}'.\phi[\vec{x}'/\vec{x}]\}.$$

- (b) We define the *cartesian* (resp. *regular*, *coherent*, *first-order*) *syntactic category* $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ (resp. $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$, $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$, $\mathcal{C}_{\mathbb{T}}^{\text{fo}}$) of a cartesian (resp. regular, coherent, first-order) theory $\mathbb{T}$ by replacing the word ‘geometric’ with ‘ $\mathbb{T}$ -cartesian’ (resp. ‘regular’, ‘coherent’, ‘first-order’) in Definition 1.4.1(a).
- (c) We define the *classical syntactic category* $\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}$ of a first-order theory $\mathbb{T}$ by replacing, in Definition 1.4.1(a), the word ‘geometric’ with ‘first-order’ and the notion of (intuitionistic) provability with that of provability in classical first-order logic.
- (d) Given a universal Horn theory $\mathbb{T}$, we define the *algebraic syntactic category* $\mathcal{C}_{\mathbb{T}}^{\text{alg}}$ of $\mathbb{T}$ as the category whose objects are the finite conjunctions of atomic formulae-in-context (up to α -equivalence) over the signature of $\mathbb{T}$ and whose arrows $\{\vec{x}.\phi\} \rightarrow \{\vec{y}.\psi\}$ (where the contexts $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_m)$ are supposed to be disjoint, without loss of generality) are sequences of terms $\vec{t} = (t_1(\vec{x}), \dots, t_m(\vec{x}))$ such that the sequent $(\phi \vdash_{\vec{x}} \psi(t_1(\vec{x}), \dots, t_m(\vec{x})))$ is provable in $\mathbb{T}$, modulo the equivalence relation which identifies two such sequences $\vec{t}$ and $\vec{t}'$ precisely when the sequent $(\phi \vdash_{\vec{x}} t_1(\vec{x}) = t'_1(\vec{x}) \wedge \dots \wedge t_m(\vec{x}) = t'_m(\vec{x}))$ is provable in $\mathbb{T}$.
- (e) We shall say that two geometric formulae-in-context $\{\vec{x}.\phi\}$ and $\{\vec{y}.\psi\}$, where $\vec{x}$ and $\vec{y}$ are disjoint, are $\mathbb{T}$ -*equivalent* if they are isomorphic as objects of the syntactic category $\mathcal{C}_{\mathbb{T}}$, that is if there exists a geometric formula $\theta(\vec{x}, \vec{y})$ which is $\mathbb{T}$ -provably functional from $\{\vec{x}.\phi\}$ to $\{\vec{y}.\psi\}$ and which moreover satisfies the property that the sequents $(\theta \wedge \theta[\vec{x}'/\vec{x}] \vdash_{\vec{x}, \vec{x}', \vec{y}} \vec{x} = \vec{x}')$ and $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta)$ are provable in $\mathbb{T}$.
- (f) We shall say that two geometric formulae $\{\vec{x}.\phi\}$ and $\{\vec{x}.\psi\}$ in the same context are $\mathbb{T}$ -*provably equivalent*, and we write $\phi \dashv\vdash_{\mathbb{T}} \psi$, if both the sequents $(\phi \vdash_{\vec{x}} \psi)$ and $(\psi \vdash_{\vec{x}} \phi)$ are provable in $\mathbb{T}$.

Lemma 1.4.2 (Lemma D1.4.4(iv) [55]). *Any subobject of $\{\vec{x}.\phi\}$ in $\mathcal{C}_{\mathbb{T}}$ is isomorphic to one of the form*

$$\{\vec{x}'.\psi[\vec{x}'/\vec{x}]\} \xrightarrow{[\psi \wedge \vec{x}' = \vec{x}]} \{\vec{x}.\phi\},$$

where $\psi(\vec{x})$ is a geometric formula in the context $\vec{x}$ such that the sequent $(\psi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}$. We will denote this subobject simply by $[\psi]$.

Moreover, for two such subobjects $[\psi]$ and $[\chi]$ we have that $[\psi] \leq [\chi]$ in $\text{Sub}_{\mathbb{C}_{\mathbb{T}}}(\{\vec{x} \cdot \phi\})$ if and only if the sequent $(\psi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$.

Theorem 1.4.3

- (i) For any cartesian theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}^{\text{cart}}$ is a cartesian category.
- (ii) For any regular theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}^{\text{reg}}$ is a regular category.
- (iii) For any coherent theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}^{\text{coh}}$ is a coherent category.
- (iv) For any first-order theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}^{\text{fo}}$ is a Heyting category.
- (v) For any geometric theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}$ is a geometric category.
- (vi) For any first-order theory $\mathbb{T}$, $\mathbb{C}_{\mathbb{T}}^{\text{fo-cl}}$ is a Boolean coherent category.

Sketch of proof The structure of subobject lattices in syntactic categories directly reflects the structure of the formulae in the relevant fragment of logic, via the identification provided by Lemma 1.4.2; for instance, unions of subobjects correspond to disjunctions, intersections to conjunctions, top element of a lattice $\text{Sub}(\{\vec{x} \cdot \phi\})$ to the associated formula $\{\vec{x} \cdot \phi\}$ (regarded as the identical subobject $[\phi]$) and bottom element of a lattice $\text{Sub}(\{\vec{x} \cdot \phi\})$, if it exists, to the formula $\{\vec{x} \cdot \perp\}$. Pullbacks correspond in a natural way to substitutions, and cover-mono factorizations of morphisms to existential quantifications (the cover-mono factorization of a morphism $[\theta(\vec{x}, \vec{y})] : \{\vec{x} \cdot \phi\} \rightarrow \{\vec{y} \cdot \psi\}$ is given by the canonical arrows $\{\vec{x} \cdot \phi\} \twoheadrightarrow \{\vec{y} \cdot (\exists \vec{x})\theta(\vec{x}, \vec{y})\} \twoheadrightarrow \{\vec{y} \cdot \psi\}$). For more details, we refer the reader to the proof of Lemma D1.4.10 [55]. $\square$

Definition 1.4.4 Let $\mathbb{T}$ be a geometric theory over a signature Σ . The *universal model* of $\mathbb{T}$ in $\mathbb{C}_{\mathbb{T}}$ is the structure $M_{\mathbb{T}}$ which assigns

- to a sort A the object $\{x^A \cdot \top\}$ where x^A is a variable of sort A ;
- to a function symbol $f : A_1 \cdots A_n \rightarrow B$ the morphism

$$\{x_1^{A_1}, \dots, x_n^{A_n} \cdot \top\} \xrightarrow{[f(x_1^{A_1}, \dots, x_n^{A_n}) = y^B]} \{y^B \cdot \top\};$$

- to a relation symbol $R \twoheadrightarrow A_1 \cdots A_n$ the subobject

$$\{x_1^{A_1}, \dots, x_n^{A_n} \cdot R(x_1^{A_1}, \dots, x_n^{A_n})\} \xrightarrow{[R(x_1^{A_1}, \dots, x_n^{A_n})]} \{x_1^{A_1}, \dots, x_n^{A_n} \cdot \top\}.$$

The attribute ‘universal’ for the structure $M_{\mathbb{T}}$ is justified by Theorem 1.4.10 below.

Theorem 1.4.5 Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then

- (i) For any geometric formula-in-context $\{\vec{x} \cdot \phi\}$ over Σ , its interpretation $[[\vec{x} \cdot \phi]]_{M_{\mathbb{T}}}$ in $M_{\mathbb{T}}$ is the subobject $[\phi] : \{\vec{x} \cdot \phi\} \twoheadrightarrow \{\vec{x} \cdot \top\}$.
- (ii) A geometric sequent $(\phi \vdash_{\vec{x}} \psi)$ over Σ is satisfied in $M_{\mathbb{T}}$ if and only if it is provable in $\mathbb{T}$.

Sketch of proof The first part of the theorem can be easily proved by induction on the structure of ϕ , while the second follows from the first by using Lemma 1.4.2. $\square$

Remark 1.4.6 Definition 1.4.4 and Theorem 1.4.5 admit obvious variants for the other fragments of first-order logic, namely cartesian, regular, coherent, intuitionistic and classical first-order logic.

1.4.2 Syntactic sites

The idea behind the notion of syntactic site is that the property of a functor to preserve the logical structure on a given syntactic category can be interpreted as a form of continuity with respect to a Grothendieck topology naturally defined on it. Before introducing such topologies, a few remarks are in order.

Recall that every arrow in a regular category can be factored uniquely as a cover followed by a monomorphism, and that covers are precisely the arrows g such that $\text{Im}(g) = 1_{\text{cod}(g)}$. In a coherent (resp. geometric) category, a finite (resp. small) *covering family* is a family of arrows such that the union of their images is the maximal subobject.

- Definition 1.4.7** (a) For a regular category $\mathcal{C}$, the *regular topology* $J_{\mathcal{C}}^{\text{reg}}$ is the Grothendieck topology on $\mathcal{C}$ whose covering sieves are those which contain a cover.
- (b) For a coherent category $\mathcal{C}$, the *coherent topology* $J_{\mathcal{C}}^{\text{coh}}$ is the Grothendieck topology on $\mathcal{C}$ whose covering sieves are those which contain finite covering families.
- (c) For a geometric category $\mathcal{C}$, the *geometric topology* $J_{\mathcal{C}}^{\text{geom}}$ is the Grothendieck topology on $\mathcal{C}$ whose covering sieves are those which contain small covering families.

Proposition 1.4.8 Let $\mathcal{C}$ and $\mathcal{D}$ be regular (resp. coherent, geometric) categories. Then a cartesian functor is regular (resp. coherent, geometric) if and only if it sends $J_{\mathcal{C}}^{\text{reg}}$ -covering (resp. $J_{\mathcal{C}}^{\text{coh}}$ -covering, $J_{\mathcal{C}}^{\text{geom}}$ -covering) sieves to covering families.

Remark 1.4.9 The Grothendieck topologies $J_{\mathcal{C}}^{\text{reg}}$, $J_{\mathcal{C}}^{\text{coh}}$ and $J_{\mathcal{C}}^{\text{geom}}$ are all subcanonical, whence for any regular (resp. coherent, geometric) category we have a Yoneda embedding $y^{\text{reg}} : \mathcal{C} \hookrightarrow \mathbf{Sh}(\mathcal{C}, J_{\mathcal{C}}^{\text{reg}})$ (resp. $y^{\text{coh}} : \mathcal{C} \hookrightarrow \mathbf{Sh}(\mathcal{C}, J_{\mathcal{C}}^{\text{coh}})$, $y^{\text{geom}} : \mathcal{C} \hookrightarrow \mathbf{Sh}(\mathcal{C}, J_{\mathcal{C}}^{\text{geom}})$).

We shall denote by $J_{\mathbb{T}}^{\text{reg}}$ (resp. by $J_{\mathbb{T}}^{\text{coh}}$, by $J_{\mathbb{T}}$) the regular (resp. coherent, geometric) topology on the regular (resp. coherent, geometric) category $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$ (resp. $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$, $\mathcal{C}_{\mathbb{T}}$) of a regular (resp. coherent, geometric) theory $\mathbb{T}$ and refer to them as the *syntactic topologies* on the relevant syntactic categories.

We shall denote by $\mathbf{Cart}(\mathcal{C}_{\mathbb{T}}^{\text{cart}}, \mathcal{D})$ (resp. $\mathbf{Reg}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, \mathcal{D})$, $\mathbf{Coh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, \mathcal{D})$, $\mathbf{Geom}(\mathcal{C}_{\mathbb{T}}, \mathcal{D})$, $\mathbf{Heyt}(\mathcal{C}_{\mathbb{T}}^{\text{fo}}, \mathcal{D})$, $\mathbf{Bool}(\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}, \mathcal{D})$) the category of cartesian (resp. regular, coherent, geometric, Heyting, coherent) functors from $\mathcal{C}_{\mathbb{T}}^{\text{cart}}$ (resp. $\mathcal{C}_{\mathbb{T}}^{\text{reg}}$, $\mathcal{C}_{\mathbb{T}}^{\text{coh}}$, $\mathcal{C}_{\mathbb{T}}$, $\mathcal{C}_{\mathbb{T}}^{\text{fo}}$, $\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}$) to a cartesian (resp. regular, coherent, geometric, Heyting, Boolean coherent) category $\mathcal{D}$.

1.4.3 Models as functors

The importance of syntactic categories lies in the fact that they allow us to associate with a theory (in the sense of axiomatic presentation), which is a ‘linguistic’, unstructured kind of entity, a well-structured mathematical object whose ‘geometry’ embodies the syntactic aspects of the theory. A most notable fact is that the models of the theory can be recovered as functors defined on its syntactic category which respect the ‘logical’ structure on it. More specifically, we have the following

Theorem 1.4.10

- (i) For any cartesian theory $\mathbb{T}$ and cartesian category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Cart}(\mathcal{C}_{\mathbb{T}}^{\text{cart}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}(\mathcal{D})$ natural in $\mathcal{D}$.
- (ii) For any regular theory $\mathbb{T}$ and regular category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Reg}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}(\mathcal{D})$ natural in $\mathcal{D}$.
- (iii) For any coherent theory $\mathbb{T}$ and coherent category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Coh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}(\mathcal{D})$ natural in $\mathcal{D}$.
- (iv) For any geometric theory $\mathbb{T}$ and geometric category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Geom}(\mathcal{C}_{\mathbb{T}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}(\mathcal{D})$ natural in $\mathcal{D}$.
- (v) For any finitary first-order theory $\mathbb{T}$ and Heyting category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Heyt}(\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}_e(\mathcal{D})$ natural in $\mathcal{D}$.
- (vi) For any finitary first-order theory $\mathbb{T}$ and Boolean category $\mathcal{D}$, we have an equivalence of categories $\mathbf{Bool}(\mathcal{C}_{\mathbb{T}}^{\text{fo-cl}}, \mathcal{D}) \simeq \mathbb{T}\text{-mod}_e(\mathcal{D})$ natural in $\mathcal{D}$.

Sketch of proof One half of the equivalence sends a model $M \in \mathbb{T}\text{-mod}(\mathcal{D})$ to the functor $F_M : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{D}$ assigning to a formula $\{\vec{x}. \phi\}$ (the domain of) its interpretation $[[\vec{x}. \phi]]_M$ in M .

The other half of the equivalence sends a functor $F : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{D}$ to the image $F(M_{\mathbb{T}})$ of the universal model $M_{\mathbb{T}}$ of $\mathbb{T}$ (cf. Definition 1.4.4) under F .

For more details, see the proof of Theorem D1.4.7 [55]. $\square$

Remark 1.4.11 By Theorem 1.4.3, the properties of Theorem 1.4.10 characterize syntactic categories up to equivalence, as representing objects for the (2-)functors $\mathcal{D} \rightarrow \mathbb{T}\text{-mod}(\mathcal{D})$ (resp. $\mathcal{D} \rightarrow \mathbb{T}\text{-mod}_e(\mathcal{D})$ in the case of intuitionistic and classical first-order syntactic categories).

The concept of syntactic category of a first-order theory also allows us to formalize the idea of a ‘dictionary’ between two theories providing a means of translating formulae in the language of the former into formulae in the language of the latter in such a way as to induce functors (resp. an equivalence) between their categories of models (inside categories possessing the required categorical structure). This is realized by the notions of interpretation (resp. bi-interpretation) of one theory into another.

Definition 1.4.12 Within a given fragment of first-order logic, an *interpretation* (resp. a *bi-interpretation*) of a theory $\mathbb{T}$ into a theory $\mathbb{S}$ is a functor (resp.

an equivalence) between their respective syntactic categories which respects the logical structure on them.

Notice that any interpretation $I : \mathcal{C}_{\mathbb{T}} \rightarrow \mathcal{C}_{\mathbb{S}}$ induces a functor $s_I : \mathbb{S}\text{-mod}(\mathcal{D}) \rightarrow \mathbb{T}\text{-mod}(\mathcal{D})$, for any ‘logical’ category $\mathcal{D}$ of the appropriate kind. However, it is not true that every functor $\mathbb{S}\text{-mod}(\mathcal{D}) \rightarrow \mathbb{T}\text{-mod}(\mathcal{D})$ is induced by an interpretation.

1.4.4 Categories with ‘logical structure’ as syntactic categories

Theorem 1.4.13 *Any small cartesian (resp. regular, coherent, geometric) category is, up to categorical equivalence, the cartesian (resp. regular, coherent, geometric) syntactic category of some cartesian (resp. regular, coherent, geometric) theory.*

Proof Let $\mathbb{T}^{\mathcal{C}}$ be the theory over the internal language $\Sigma_{\mathcal{C}}^f$ of $\mathcal{C}$ (cf. section 1.3.1.1) consisting of the following sequents:

$$(\top \vdash_x (\ulcorner f \urcorner(x) = x))$$

for any identity arrow f in $\mathcal{C}$;

$$(\top \vdash_x (\ulcorner f \urcorner(x) = \ulcorner h \urcorner(\ulcorner g \urcorner(x))))$$

for any triple of arrows f, g, h of $\mathcal{C}$ such that f is equal to the composite $h \circ g$;

$$(\top \vdash_{\square} (\exists x)\top) \text{ and } (\top \vdash_{x,x'} (x = x'))$$

where x and x' are of sort $\ulcorner 1 \urcorner$, 1 being the terminal object of $\mathcal{C}$;

$$\begin{aligned} &(\top \vdash_x (\ulcorner h \urcorner(\ulcorner f \urcorner(x)) = \ulcorner k \urcorner(\ulcorner g \urcorner(x)))), \\ &(((\ulcorner f \urcorner(x) = \ulcorner f \urcorner(x')) \wedge (\ulcorner g \urcorner(x) = \ulcorner g \urcorner(x')))) \vdash_{x,x'} (x = x')), \text{ and} \\ &((\ulcorner h \urcorner(y) = \ulcorner k \urcorner(z)) \vdash_{y,z} (\exists x)((\ulcorner f \urcorner(x) = y) \wedge (\ulcorner g \urcorner(x) = z))) \end{aligned}$$

for any pullback square

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ \downarrow g & & \downarrow h \\ c & \xrightarrow{k} & d \end{array}$$

in $\mathcal{C}$.

It is immediate to see that for any cartesian category $\mathcal{D}$, $\mathbb{T}^{\mathcal{C}}$ -models are ‘the same thing’ as cartesian functors $\mathcal{C} \rightarrow \mathcal{D}$. So we have an equivalence of categories $\mathbb{T}^{\mathcal{C}}\text{-mod}(\mathcal{D}) \simeq \mathbf{Cart}(\mathcal{C}, \mathcal{D})$ natural in $\mathcal{D} \in \mathbf{Cart}$. Since we also have an equivalence $\mathbf{Cart}(\mathcal{C}_{\mathbb{T}^{\mathcal{C}}}^{\text{cart}}, \mathcal{D}) \simeq \mathbb{T}^{\mathcal{C}}\text{-mod}(\mathcal{D})$ natural in $\mathcal{D} \in \mathbf{Cart}$ (by Theorem 1.4.10), by composing the two we find an equivalence $\mathbf{Cart}(\mathcal{C}, \mathcal{D}) \simeq \mathbf{Cart}(\mathcal{C}_{\mathbb{T}^{\mathcal{C}}}^{\text{cart}}, \mathcal{D})$ natural in $\mathcal{D} \in \mathbf{Cart}$ and hence, by the (2-dimensional) Yoneda lemma, an equivalence of categories $\mathcal{C}_{\mathbb{T}^{\mathcal{C}}}^{\text{cart}} \simeq \mathcal{C}$, one half of which sends a formula $\phi(\vec{x})$ to (the domain of) its interpretation $[[\vec{x}. \phi]]$ in the tautological $\Sigma_{\mathcal{C}}^f$ -structure in $\mathcal{C}$.

Given a Grothendieck topology J on a cartesian category $\mathcal{C}$, let us denote by $\mathbb{T}_J^{\mathcal{C}}$ the theory obtained from $\mathbb{T}^{\mathcal{C}}$ by adding all the sequents of the form

$$\left(\top \vdash_x \bigvee_{i \in I} (\exists y_i) (\ulcorner f_i \urcorner (y_i) = x) \right)$$

for a J -covering family $\{f_i : B_i \rightarrow A \mid i \in I\}$. By Proposition 1.4.8, if $\mathcal{C}$ is a regular (resp. coherent, geometric) category then for any regular (resp. coherent, geometric) category $\mathcal{D}$ the regular (resp. coherent, geometric) functors $\mathcal{C} \rightarrow \mathcal{D}$ are exactly the cartesian functors on $\mathcal{C}$ which are J -continuous, where J is the regular (resp. coherent, geometric) topology on $\mathcal{C}$. So we conclude as above that if $\mathcal{C}$ is a regular (resp. coherent, geometric) category then there is an equivalence of categories $\mathcal{C}_{\mathbb{T}_J^{\mathcal{C}}}^{\text{reg}} \simeq \mathcal{C}$ (resp. $\mathcal{C}_{\mathbb{T}_J^{\mathcal{C}}}^{\text{coh}} \simeq \mathcal{C}$, $\mathcal{C}_{\mathbb{T}_J^{\mathcal{C}}}^{\text{geom}} \simeq \mathcal{C}$) one half of which sends a formula $\phi(\vec{x})$ to (the domain of) its interpretation $[[\vec{x}. \phi]]$ in the canonical $\Sigma_{\mathcal{C}}^{\text{f}}$ -structure in $\mathcal{C}$. $\square$

Remark 1.4.14 The fact that every cartesian (resp. regular, coherent, geometric) category $\mathcal{C}$ is equivalent to the syntactic category of a theory $\mathbb{T}$ allows one to interpret categorical constructions on $\mathcal{C}$ as logical operations in $\mathbb{T}$.

1.4.5 Soundness and completeness

Theorem 1.4.15 (Soundness). *Let $\mathbb{T}$ be a cartesian (resp. regular, coherent, geometric, first-order) theory over a signature $\mathbb{T}$, and let M be a model of $\mathbb{T}$ in a cartesian (resp. regular, coherent, geometric, Heyting) category $\mathcal{C}$. If σ is a sequent (in the appropriate fragment of first-order logic over Σ) which is provable in $\mathbb{T}$, then σ is satisfied in M .*

Proof The thesis immediately follows from Theorems 1.4.5 and 1.4.10 in view of Remark 1.4.6. $\square$

Theorem 1.4.16 (Strong completeness). *Let $\mathbb{T}$ be a cartesian (resp. regular, coherent, geometric, first-order) theory. If a cartesian (resp. regular, coherent, geometric, Heyting,) sequent σ is satisfied in all models of $\mathbb{T}$ in cartesian (resp. regular, coherent, geometric, Heyting) categories, then it is provable in $\mathbb{T}$.*

Proof Thanks to the construction of syntactic categories, the property of completeness with respect to all the models in categories of the appropriate kind becomes a tautology (cf. Theorem 1.4.5). $\square$

Let us now discuss classical completeness, that is, completeness with respect to the class of set-based models.

Definition 1.4.17 Within a given fragment of geometric logic, a theory $\mathbb{T}$ is said to have *enough set-based models* if for any geometric sequent σ over its signature which is valid in all the set-based models of $\mathbb{T}$, σ is provable in $\mathbb{T}$.

The following theorem is equivalent, via the notion of Morleyization of a first-order theory discussed in section 2.1.1, to Gödel's classical completeness theorem for classical first-order logic.

Theorem 1.4.18 (Classical completeness for coherent logic, cf. Corollary D1.5.10(ii) [55]). *Assuming the axiom of choice, every coherent theory has enough set-based models.*

Remark 1.4.19 Having extended the notion of model from **Set** to an arbitrary 'logical' category, it is no longer necessary, as in classical finitary first-order logic, to appeal to non-constructive principles such as the axiom of choice to ensure the existence of 'enough' models of the theory (that is, of a class of models such that validity in all of them amounts to provability in the theory). Indeed, the universal model $M_{\mathbb{T}}$ in the syntactic category of $\mathbb{T}$ exists constructively and enjoys a strong form of completeness (cf. Theorem 1.4.5): it is literally made up of formulae, and represents a natural first-order analogue of the Lindenbaum-Tarski algebra of a propositional theory (cf. Remark 2.1.20 below).